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OF ICELAND**

Wind, Temperature, and Rural Injury Accidents in Iceland

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Wind, Temperature, and Rural Injury Accidents in Iceland

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**30 ECTS thesis submitted in partial fulfilment of a
Master of Science degree in Software Engineering**

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Abstract

This study examines whether rural injury accidents in Iceland occur disproportionately during strong wind. Accident records from 2007–2025 were linked to nearby weather observations. The primary analysis compared observed accident counts with those expected from the local frequency of each weather condition. Two further analyses examined the role of traffic: an approximate adjustment for weather-related changes in traffic, and an analysis of accident rates using estimated vehicle-kilometres travelled (VKT), derived from daily traffic counts and road length. Mean wind was the primary weather measure; gust and temperature provided complementary comparisons.

Strong wind was over-represented at accident times. At mean wind speeds of at least 20 m/s, the observed-to-expected accident ratio (O/E) was 2.15. Counters showed lower traffic in strong wind. Accounting approximately for this reduction strengthened rather than removed this pattern. The VKT analysis also showed higher rates in the strongest wind intervals, although those intervals contained few accidents. Gust showed a similar pattern. Temperature had a non-linear association: O/E was elevated just below freezing and at the warm end, while VKT rates were highest in cold conditions and rose again above the 9–12°C minimum. The warm-end VKT increase was smaller than the O/E increase.

The findings support an association between strong wind and rural injury accidents, not a causal effect. Weather observations cannot fully describe conditions on the road, and the traffic analyses rely on selective counter coverage and estimated exposure within each day. Hourly traffic data with broader geographic coverage would be needed for a direct national estimate of accident risk per distance travelled.

Ágrip

Í þessari rannsókn er skoðað hvernig umferðarslys með meiðslum í dreifbýli á Íslandi tengjast veðri á árunum 2007–2025, með sérstakri áherslu á mikinn vind. Slys voru tengd við veðurmælingar frá nálægum stöðvum og fyrst kannað hvort slys væru hlutfallslega algengari við ákveðin veðurskilyrði en búast mætti við miðað við hversu oft þau skilyrði komu fyrir á hverjum stað. Umferð var síðan tekin inn í greininguna með tvennum hætti: annars vegar með nálgun sem leiðréttir fyrir breytingum á daglegri umferð í slæmu veðri og hins vegar með samanburði á slysátíðni miðað við áætlaðan akstur á vegum þar sem umferðarteljarar voru til staðar. Meðalvindur var meginveðurbreytan, en vindhviður og hitastig voru einnig skoðuð.

Mikill vindur var oftast til staðar við slys en vænta mátti út frá tíðni hans. Við meðalvindhraða 20 m/s eða meiri var fjöldi skráðra slysa 2,15-faldur miðað við þann fjölda sem vænta mátti út frá tíðni slíks vinds. Umferð samkvæmt teljuragögnum dróst saman í miklum vindi og þegar gróflega var leiðrétt fyrir því jókst hlutfall skráðra og væntanlegra slysa enn frekar. Greining byggð á áætluðum eignum kílómetrum sýndi einnig hærri slysátíðni við mesta vindinn, en fá slys lágu að baki þeim niðurstöðum. Vindhviður sýndu svipað mynstur. Tengsl hitastigs við slys voru ólínuleg: slys voru hlutfallslega algengari rétt undir frostmarki og við hæsta hitastigið miðað við veðurtíðni. Slysátíðni á hvern áætlaðan ekinn kílómetra var hæst í kulda og jókst aftur við hæsta hitastigið miðað við lágmarkið á bilinu 9–12 gráður. Sú aukning var þó minni en í samanburðinum við veðurtíðni.

Niðurstöðurnar benda því til tengsla milli mikils vinds og slysa með meiðslum í dreifbýli, en sýna ekki að vindur valdi slysum. Veðurmæling frá einni stöð lýsir ekki fullkomlega aðstæðum á vegi og umferðargögnin ná aðeins til hluta vegakerfisins og liggja fyrir sem daglegar tölur en ekki eftir klukkustundum. Til að meta slysaáhættu á hvern ekinn kílómetra fyrir landið allt þyrfti víðtækari umferðarmælingar og gögn eftir klukkustundum.

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Abbreviations

O/E	Observed-to-expected accident ratio.
VKT	Vehicle-kilometres travelled.

1 Introduction

1.1 Context and motivation

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are relevant to Icelandic roads, where terrain can produce substantial local variation in wind (Snæbjörnsson et al., 2007; Pu et al., 2018). Precipitation and fog are also relevant road-weather hazards, but they are context for this study and are not analysed here.

Previous research provides two reasons to examine wind. Engineering studies describe mechanisms by which crosswind can destabilise vehicles, while observational studies compare accident occurrence with the frequency of windy conditions (Baker, 1986, 1994; Edwards, 1994). These are complementary forms of evidence: a physical mechanism does not establish how often accidents occur across a road network. Reviews also find less consistent road-safety evidence for wind than for precipitation (Theofilatos and Yannis, 2014). Local evidence is therefore needed to determine whether the same pattern is present on Icelandic roads.

In Iceland, wind-related road safety has been studied using vehicle-reliability analyses and operational vehicle-wind models (Snæbjörnsson et al., 2007; Hrafnkelsdóttir, 2020). This thesis addresses a different question: whether rural injury accidents are over-represented under particular weather conditions relative to how often those conditions occur locally. It then examines how the interpretation changes when available traffic information is introduced. The study therefore links a long Icelandic accident series with local weather frequency and two complementary traffic-exposure approaches, while treating station observations as proxies for road conditions rather than direct measurements of the forces acting on a vehicle.

The study covers rural injury accidents in Iceland from 2007–2025. Accident times and locations are linked to nearby 10-minute weather observations. Mean wind is the primary weather measure; wind gust and temperature are secondary measures, while season, time of day and daylight provide additional context.

Accident counts alone do not measure accident risk. Severe weather may increase the hazard faced by each vehicle while also reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately (Andrey et al., 2003; Lord and Mannering, 2010; Washington et al., 2020).

1.2 Aim and analysis strategy

The primary research question is: *How does rural injury accident occurrence vary with weather, particularly strong wind, after accounting as far as possible for local weather frequency and traffic exposure?*

Three steps guide the analysis:

1. **Weather-frequency O/E analysis.** Compare observed accidents with the number expected from local weather frequency, by parameter and season. This measures over-representation relative to weather time, not risk per vehicle.
2. **Approximate traffic-corrected O/E analysis.** Establish how traffic changes with weather, then use the observed counter response to reweight the baseline comparison. This asks whether weather-related changes in traffic explain the observed weather patterns, particularly the high-wind pattern.
3. **Vehicle-kilometres travelled (VKT) analysis.** Relate accident counts to estimated vehicle-kilometres using daily traffic, rural road length and monthly weather frequencies, including eligible accident-free counter-days.

The three analyses use different denominators and, for the VKT analysis, a different study sample. The baseline and approximate correction cover 2007–2025, whereas the VKT analysis uses a smaller 2019–2024 sample. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

1.3 Thesis outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, data preparation, and statistical methods. Chapter 4 presents the three main analyses in the order of these steps, followed by interpretation of their differences. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions.

2 Background

2.1 Wind and rural road accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone (Baker, 1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study by Snæbjörnsson et al. (2007) modelled accident probability as a combined function of wind velocity and direction, road camber and friction, and vehicle speed. A station measures wind at one point, whereas vehicle stability also depends on the wind's direction relative to the road and on local road conditions. Terrain between a station and a road can alter the wind reaching the vehicle. Station observations therefore remain exposure proxies, and a shorter matching distance does not guarantee identical conditions. Mobile road measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve (Pu et al., 2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars (Hrafnkelsdóttir, 2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here (Edwards, 1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error (Young and Liesman, 2007). More broadly, Theofilatos and Yannis (2014) report less consistent road-safety evidence for wind than for precipitation.

2.2 Temperature, daylight, and time of day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series study found its strongest cold-weather association below freezing in winter, while an hourly study in Shenzhen found a J-shaped temperature relationship (Lee et al., 2014; Zhan et al., 2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and

Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk (Johansson et al., 2009). Here, raw hour, season, and daylight counts describe the sample. Without hourly or daylight-specific traffic exposure, these counts cannot identify differences in accident risk.

2.3 Traffic and accident rates

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume (Sathiaraj et al., 2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle is higher. High-resolution crash studies illustrate the value of hourly weather data rather than daily or monthly averages (Xing et al., 2019). The analyses distinguish three quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic.

These studies show why weather frequency and traffic exposure need to be considered separately when interpreting accident counts. The analyses therefore begin with local weather-frequency standardisation and then introduce traffic through an approximate correction and a VKT analysis.

3 Data and methods

3.1 Data

3.1.1 Source data

The data were obtained from four government institutions: the Icelandic Transport Authority (*Samgöngustofa*; ITA), the Icelandic Road and Coastal Administration (*Vegagerðin*; IRCA), the Icelandic Meteorological Office (*Veðurstofa Íslands*; IMO), and Statistics Iceland (*Hagstofa Íslands*); see Table 3.1.

Table 3.1: Source datasets and coverage.

Data	Provider	Contents	Coverage	Reference
Accidents	ITA	Accident events	126,607 events, 2007–2025	ITA (2026a)
Weather	IMO	10-minute observations	232,459,562 observations; 338 stations, 2007–2025	IMO (2026)
Traffic/roads	IRCA	Seasonal averages, daily counts and road geometry	1,226 mapped road sections; 33,757 section-year records of seasonal averages, 2000–2025; 1,033,659 daily channel records, 2019–2024	IRCA (2026)
Urban areas	Statistics Iceland	Urban boundaries	97 polygons, 2020–2024 release	Statistics Iceland (2026)

Records from the year 2025 are included in the accident, injury, vehicle, road, seasonal-traffic and weather datasets under the same selection and quality criteria as records from earlier years. Daily counter data are available through 2024. Table 3.2 defines the weather and traffic quantities used in the subsequent comparisons.

Table 3.2: Definitions used throughout the thesis.

Term	Definition	Use in this thesis
f	10-minute mean wind speed, m/s	Main weather measure, matched within five minutes of the accident time.
fg	Wind gust reported in the same 10-minute observation, m/s	Additional weather measure; not a daily maximum.
t	Air temperature reported with each 10-minute observation, °C	Additional weather measure matched independently.
O/E	Observed accidents divided by accidents expected from local weather frequency	Main relative-occurrence result; traffic is not included.

3.1.2 Accident data

The accident dataset contains 126,607 records from 2007–2025 (ITA, 2026a). These events come from the official register, which is based primarily on police reports (ITA, 2026b). One record lacks valid temporal or spatial information, leaving 126,606 valid records.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons (Statistics Iceland, 2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

Applying this rural classification yields 24,005 accidents, of which 6,414 involve injury across the rural road network (Figure 3.1). Of these, 6,259 have a qualifying primary weather match within 20 km and five minutes.

The selection follows the scientific focus: rural roads may be more directly exposed to local weather conditions, and rural records contain a larger injury share. Of 102,601 urban accidents, 10,631 involved injury (10.4%), compared with 6,414 of 24,005 rural accidents (26.7%). Among injury accidents, serious or fatal outcomes accounted for 1,802 urban records (17.0%) and 1,471 rural records (22.9%). These are recorded counts and proportions, not exposure-adjusted accident rates. The analysis first restricts to rural injury accidents, separates minor from serious or fatal injury, and then requires qualifying nearby weather. The maximum distance of 20 km is a pragmatic coverage/proximity compromise, not a physical safety threshold.

Accident severity is recorded in `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is `meidsli ≤ 3`. The serious or fatal comparison uses `meidsli ≤ 2`. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

For the reader-facing results, injury accidents are divided into two disjoint groups: minor injury (`meidsli=3`) and serious or fatal injury (`meidsli=1` or `2`). The O/E calculation for these groups is described in Section 3.3.1.

Table 3.3 shows one cleaned rural injury accident record before weather matching.

Table 3.3: Example of a cleaned rural injury accident record. The table illustrates accident-level information retained before weather matching. The source record identifier and all displayed fields are reproduced from that same cleaned accident record.

Field	Example value
Source record identifier	3168111
Date	2007-01-03
Recorded time	18:08:00
Registered road section	1-g8
Latitude / longitude (degrees)	64.57709888007743, -21.879289941375845
Area classification	Rural
Injury category	Minor injury
Involved vehicles	1

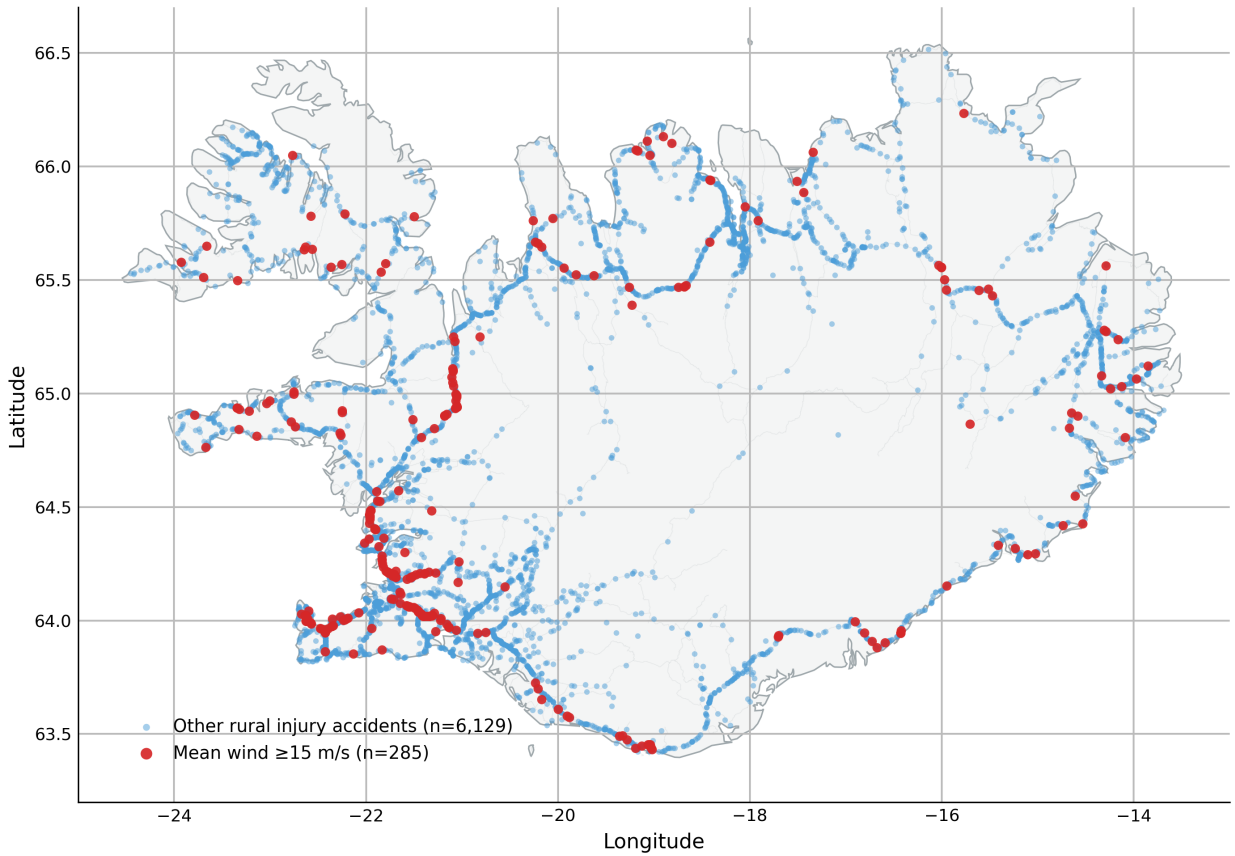


Figure 3.1: Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study. Accidents follow the main rural road network.

3.1.3 Weather data

The weather dataset contains 10-minute observations from Icelandic weather stations. The variables used here are 10-minute mean wind speed (f), reported wind gust (fg), and temperature (t). For each accident, the gust is taken from the same matched 10-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

$$0 \leq f < 45, \quad 0 \leq fg < 65, \quad fg + 0.5 \geq f, \quad (3.1)$$

where f and fg are mean wind speed and gust, respectively, in m/s.

Rows missing either f or fg , values outside these bounds, and internally inconsistent records where $fg = 0$ but $f > 0$ are excluded. A value of $f = 0$ alone is retained; only uninterrupted runs with $f = fg = 0$ lasting at least two hours are excluded as potential sensor stoppages. Table 3.5 gives the full cleaning summary. Missing temperature does not affect wind eligibility.

Each accident is matched to the geographically nearest eligible wind station within 20 km, using the closer of the two adjacent 10-minute timestamps. The selected observation is therefore within five minutes of the recorded accident time and can fall on the neighbouring calendar day for accidents close to midnight. Matching eligibility is recorded for all 6,414 rural injury accidents.

Temperature is matched independently within the same 20 km and five-minute limits. The nearest valid value between -30 and 30°C may therefore come from a different station when the wind station

lacks temperature. In the final analysis dataset, however, all 6,259 wind and temperature matches use the same station and observation time.

The same cleaned weather archive is used for accident matching, background frequencies and traffic analyses; temperature is not imputed.

Table 3.4 shows the coverage gained as the matching radius increases. A distant station is a less desirable proxy for road conditions, but there is no physically exact cut-off. The 20 km threshold was selected as a practical compromise between high coverage and avoiding very distant stations; it was not determined by an optimisation procedure.

Table 3.4: Weather-match coverage at alternative station-distance thresholds. The denominator is 6,414 rural injury accidents, 2007–2025; all matches require valid mean wind and gust within five minutes.

Maximum distance (km)	Matched accidents	Matched (%)
5	3,060	47.7%
10	5,098	79.5%
20	6,259	97.6%
30	6,402	99.8%

Table 3.5: Overview of cleaning of wind measurement records, 2007–2025. Every supplied station-year contains at least one wind measurement.

Category	Records	Share
All supplied station-time observations	232,459,562	100.00%
Missing f or fg	261,519	0.11%
$f < 0$, $fg < 0$, $f \geq 45$ or $fg \geq 65$ m/s	8,766	0.00%
$fg = 0 < f$ or $fg + 0.5 < f$ (m/s)	257,245	0.11%
All-zero runs ≥ 2 hr	1,473,082	0.63%
Wind observations retained	230,458,950	99.14%

The largest excluded category is uninterrupted all-zero runs, treated separately as possible instrument stoppages. O/E background frequencies use all retained observations in each accident’s station–season group, not only observations from accident days.

3.1.4 Traffic data

Daily counters provide the traffic information used in the main analysis. Seasonal road-section averages are also available for 2000–2025 (IRCA, 2026), and provide the coarser full-period exposure approximation used for the descriptive winter–summer and road-rate summaries. They do not replace daily counts in the main VKT weather analysis.

The study seasons follow the traffic periods: winter is December–March, spring April–May, summer June–September, and autumn October–November. These are operational, traffic-aligned seasons rather than conventional meteorological seasons; the IMO also uses June–September when summarising the Icelandic summer (IMO, 2025). Daily traffic counts from Vegagerðin cover 2019–2024 and underpin both the traffic response and counter-linked exposure calculations. They are daily, not hourly, totals. Counter positions are estimated from registered road geometry, and more than one counter can occur on a road section. These data remain separate from the primary O/E calculation.

3.1.5 Selection summary

Table 3.6 summarises the main selections. The first four rows follow accident selection sequentially; the final rows start from separate populations for the VKT analysis, weather observations, and counter-days. Exclusion does not necessarily imply an erroneous source record. The 763,171 counter-days describe the traffic-response analysis; the main VKT denominator uses 533,649 eligible counter-section days.

Table 3.6: Overview of primary data selections for accidents, weather and traffic counts. The first four rows are sequential; the last three rows start from separate populations. Each removal percentage uses the count entering that row.

Selection	Entering	Removed at step (%)	Retained	Reason
Valid accident records	126,607	< 0.01%	126,606	Invalid time or coordinates.
Rural accidents	126,606	81.04%	24,005	Inside the urban study boundary.
Rural injury accidents	24,005	73.28%	6,414	No recorded injury.
Accidents for weather-only analysis	6,414	2.42%	6,259	No qualifying weather within 20 km and five minutes.
Accidents for VKT analysis	1,863	62.75%	694	Separate 2019–2024 sample: counter linkage, daytime, weather and traffic eligibility.
Wind observations	232,459,562	0.86%	230,458,950	Wind exclusions in the cleaning overview.
Counter-days for traffic-response analysis	774,274	1.43%	763,171	Usable daytime weather summary.

3.1.6 Descriptive sample and exposure context

The seasonal exposure and conditional severity comparisons provide descriptive context, not a fourth main analysis. The three main analyses are defined in Section 3.3.

Figure 3.2 describes sample composition before exposure adjustment. Counts peak in the afternoon, and serious or fatal injury accidents are more numerous in summer than winter. These raw differences may reflect travel patterns as well as weather; they cannot be interpreted directly as risk.

Counts by hour and daylight cannot be interpreted as accident risk because the corresponding traffic exposure by hour or daylight category is unavailable. In particular, fewer night-time accidents do not imply safer night-time driving. Temperature differs because the weather archive supplies the local frequency of temperature conditions, allowing denominator-based comparisons in the weather-frequency O/E analysis (Section 4.2.3), approximate traffic-corrected O/E analysis (Section 4.3.3) and VKT analysis (Section 4.4.1).

Among injury accidents, 670 of 2,387 (28.1%) were serious or fatal in summer, compared with 384 of 2,083 (18.4%) in winter. Conditional on an injury accident, the serious/fatal proportion was therefore about 52% higher in summer. This is a summer–winter severity comparison, not summer versus the rest of the year and not an exposure-adjusted accident rate.

The descriptive full-period traffic calculation multiplies winter or summer average daily traffic by mapped rural road length and the number of winter or summer days in each calendar year. Urban portions are removed with the same boundaries as accident classification. Although winter had 12.7% fewer injury accidents, estimated driving was 54.5% lower. The estimated winter/summer rate ratios were 1.92 for all injury accidents, 2.17 for minor injuries and 1.26 for serious/fatal injuries. The seasonal rate comparison uses estimated VKT as its denominator, whereas the severity comparison

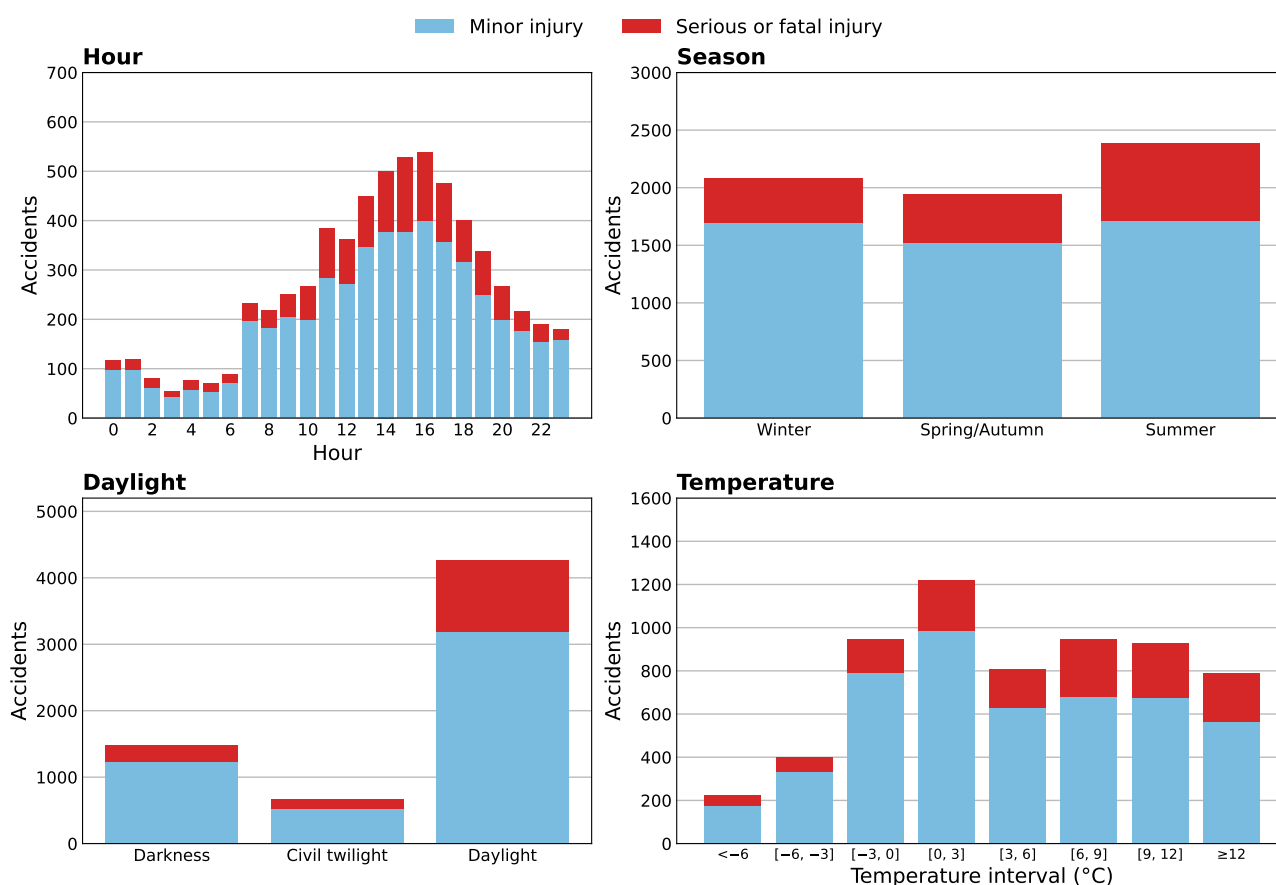


Figure 3.2: Distribution of rural injury accidents by hour, study season, estimated daylight and matched temperature. Blue shows minor injury (code 3), and red shows serious or fatal injury (codes 1–2). Spring and autumn are combined in this descriptive figure. Hour, season and daylight use 6,414 accidents; temperature uses 6,259 qualifying matches.

Table 3.7: Descriptive rural injury accidents and estimated seasonal travel, 2007–2025. Winter or summer average daily traffic is multiplied by mapped rural road length and calendar days. Unmapped portions are excluded.

Quantity	Winter	Summer
Injury accidents	2083	2387
Serious or fatal injury accidents	384	670
Estimated driving (billion vehicle-km)	6.32	13.89
Injury accidents per million estimated vehicle-km	0.329	0.172

conditions on an injury accident having occurred. The higher summer severity share and higher winter serious/fatal rate are therefore compatible.

The reproducible clipping gives 6.32 billion winter and 13.89 billion summer vehicle-kilometres. Treating every unmapped portion as rural changes the overall ratio from 1.92 to 1.91. The 2007 workbook omits station distances: origins are recovered from the nearest available records for the same section and named starting point, requiring agreement when both earlier and later records exist. Published lengths are retained; unresolved origins and unmapped geometry are excluded. These approximations include all rural injury counts but incomplete historical road exposure and may overestimate absolute rates. Section 4.4 uses observed daily counters in a smaller sample; Figure 4.14 gives its seasonal wind comparison.

3.2 Data preparation and reproducibility

The workflow separates the preparation and cleaning of data from the analysis of prepared inputs. Four institutions supply the primary data used in the study: Samgöngustofa provides data on accidents; the Meteorological Offices provides weather observations and station information; Vegagerðin provides daily counters, annual traffic summaries and road geometry; and Statistics Iceland provides urban-locality boundaries.

The preparation standardises identifiers, dates, coordinates and injury codes, applies the rural classification, and screens weather using the validity rules of Section 3.1. Road geometry and counter records establish rural counter sections and their daily traffic. Accident-time wind and temperature are linked to nearby valid observations within 20 km and five minutes. Station and time checks distinguish a successful match from an unmatched event. The primary prepared data files are shown in Table 3.8.

Table 3.8: Principal prepared data files used in the analyses. Paths are relative to `data/processed/`.

Prepared file	Contents
<code>accidents/all.csv</code>	Standardised accident records.
<code>weather/weather.parquet</code>	Cleaned ten-minute weather observations.
<code>weather/frequency.csv</code>	Weather-interval frequencies by station and season.
<code>accidents/rural_injury.csv</code>	Rural injury accidents matched to weather at the time of the accident.
<code>traffic/annual.csv</code>	Annual and seasonal traffic estimates by road section.
<code>traffic/counter_days.csv</code>	Daily traffic counts and rural vehicle-kilometres by counter-section.
<code>traffic/counter_accidents.csv</code>	Eligible accidents linked to counter-sections and weather observations.
<code>weather/monthly_frequency.csv</code>	Weather-interval frequencies by station and calendar month.
<code>traffic/traffic_weather_response.csv</code>	Estimated variation in traffic across weather intervals.

The preparation and analysis stages of the project are described in a separate file, `docs/pipeline.md`. Table 3.9 shows its first four preparation stages. Tables 3.10 and 3.11 are generated directly from the downstream cleaned files `data/processed/accidents/all.csv` and `data/processed/weather/weather.parquet`, respectively. They show the first two stored records of each file.

Table 3.9: First four preparation stages from *docs/pipeline.md*. Input and output paths are relative to *data/*; script paths are relative to *src/*.

Script	Input files	Output files	Description
accidents/ build.py	raw/accidents/: accidents_*.txt, vehicles_*.txt, road_links_2007_2025.csv, urban_boundaries_2020_2024. geojson	processed/accidents/all. csv	Joins accident, vehicle, road-link and boundary data.
weather/ download_ weather.py	Official delivery: stod.txt, f_*.txt, fj_*.txt, fv_*.txt	raw/weather/: stations.csv, weather_10min_raw. parquet, weather_10min_raw_audit. csv	Downloads and combines the complete station-file delivery without filtering.
weather/ clean.py	raw/weather/weather_10min_ raw.parquet	processed/weather/: weather.parquet, cleaning.csv	Applies the fixed weather-quality rules and records annual counts.
weather/ frequency.py	processed/weather/weather. parquet	processed/weather/: frequency.csv, yearly_frequency.csv, temperature_frequency. csv, traffic_frequency.csv	Counts station-season and station-year-season wind, gust and temperature frequencies.
:	:	:	:

Table 3.10: First two stored records from the cleaned accident dataset.

id	timestamp	lat	lon	meidsli
3166374	2007-01-01 00:38:00	64.12584278999495	-21.84899049625977	3
3166372	2007-01-01 00:44:00	64.13069955245595	-21.915262585087383	4
:	:	:	:	:

Same records, remaining columns.

tegohapps	vehicle_count	urban_rural	registered_road_section
121	2	Urban	41-04
920	1	Urban	418-01
:	:	:	:

Note: All columns are shown in source order across the two blocks; CSV values are reproduced without relabelling or rounding.

Table 3.11: First two stored records from the cleaned weather dataset.

station	time	f	fg	t
1350	2011-06-01T20:30:00	7.28	9.28	7.11
1350	2011-06-01T20:40:00	7.13	9.28	7.14
⋮	⋮	⋮	⋮	⋮

Note: Floating-point values use the shortest decimal representation that round-trips to the stored Parquet dtype.

3.3 Analysis

3.3.1 Weather-frequency O/E analysis

The observed-to-expected ratio (O/E) compares accidents observed in each weather interval with the number expected from local weather frequency. Accident-time mean wind (f) is the primary measure and is grouped in 5 m/s intervals; gust from the same 10-minute observation and independently matched temperature are analysed in the same framework. For each measure, expected counts use the weather distribution at the matched station in the same study season, pooled across study years, rather than a national distribution.

The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}, \quad (3.2)$$

where j indexes weather intervals, g identifies a station–season group, T_{jg} counts its valid 10-minute observations in interval j , T_g counts all its valid observations, and p_{jg} is the resulting weather-time proportion.

If accident timing followed local weather frequency, the expected number in an interval would be the group accident total multiplied by its weather-time proportion. The pooled observed-to-expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}, \quad (3.3)$$

where R_j is O/E for weather interval j , A_{jg} is its observed accident count in station–season group g , $A_g = \sum_j A_{jg}$ is the group accident total, and p_{jg} is its background weather proportion. The sums over g pool observed and expected counts across groups.

An O/E of one means that the accident share equals the local time share of the weather interval; values above one indicate over-representation. For example, 56 minor-injury accidents occurred at mean wind of at least 20 m/s compared with 25.18 expected, giving O/E 2.22. Standardising within station and season prevents differences in local weather distributions from dominating the pooled ratio.

Minor injury ($mei\delta sli = 3$) and serious or fatal injury ($mei\delta sli \leq 2$) are analysed as disjoint groups, each with its own station–season expected counts. Combined all-injury O/E uses summed observed and expected counts rather than averaging group ratios. Because this is a descriptive standardised ratio rather than a fitted model, the O/E figures do not show model-based confidence intervals.

Temperature was grouped into 3°C intervals from -6 to 12°C , with open-ended groups below -6°C and at or above 12°C .

The original 5 m/s intervals are the normal descriptive scale for wind. The pooled primary contrast uses the extreme mean-wind category of at least 20 m/s. The coarse joint summary and some seasonal displays combine mean winds from 15 m/s to avoid excessively sparse subdivisions; the detailed joint figure retains 15–20 and at least 20 m/s as separate groups. These grouping choices were not statistically optimised and are not universal physical danger thresholds.

The detailed joint wind–temperature figure uses mean-wind intervals $[0, 5]$, $[5, 10]$, $[10, 15]$, $[15, 20]$ and ≥ 20 m/s, and temperature intervals < -6 , $[-6, -3]$, $[-3, 0]$, $[0, 3]$, $[3, 6]$, $[6, 9]$, $[9, 12]$ and ≥ 12 °C, within the existing weather-validity limits. These are descriptive categories, not physical safety thresholds.

Computationally, each simultaneous valid wind and temperature observation enters exactly one of the 40 cells. Expected counts use the actual joint frequency within the same station–season groups as Equation 3.3, rather than products of marginal frequencies. Accident matches must share station and observation time. The 40 cells can be pooled into the coarser wind $< 15/\geq 15$ m/s and temperature $< 3/\geq 3$ °C summary; observed and expected totals each equal the eligible joint sample.

The joint description uses weather-frequency O/E because simultaneous observations and the larger 2007–2025 matched sample support a joint weather-time denominator. The 694-accident VKT sample instead uses allocated station–month exposure, not directly observed simultaneous wind–temperature travel. Cell O/E therefore describes accident occurrence relative to local weather frequency, not a vehicle-based rate. Cells with fewer than 10 observed or 5 expected accidents are marked as sparse. Descriptive contrasts divide O/E at wind ≥ 15 m/s by O/E at 0–5 m/s within temperature groups, and O/E at ≥ 12 °C or -3 – 0 °C by O/E at 6– 12 °C within wind groups. Contrasts are reported only when both cells meet the support rule; they are not relative risks or formal interaction estimates.

3.3.2 Use of traffic data

The primary O/E denominator is the local frequency of wind conditions rather than traffic volume. Complete traffic data are unavailable at the required spatial and temporal resolution, so the weather-frequency O/E analysis does not account for changes in travel volume during strong wind. If traffic is lower in strong wind while other factors remain unchanged, the accident-rate contrast per vehicle-kilometre would be larger than the weather-frequency O/E contrast.

Traffic information is used in two distinct ways. The approximate correction estimates the relative daily traffic response across weather intervals from the 2019–2024 counters and uses that response to reweight the weather-frequency O/E denominator. The VKT analysis uses daily traffic and rural road length to estimate exposure in a smaller counter-linked sample; its allocation across weather intervals is described in Section 3.3.4. Neither analysis observes hourly traffic.

3.3.3 Approximate traffic-corrected O/E analysis

The approximate correction uses the observed day-to-day traffic response at counter sections during 2019–2024. Among positive-count days, the mean count for the same counter section, year, month, and weekday defines expected daily traffic. Each day’s observed 24-hour count and expected count are allocated to weather intervals in proportion to their valid 07:00–24:00 weather minutes on that date. For interval j , r_j is the sum of allocated observed traffic divided by the sum of allocated expected traffic. Zero-count days are excluded from this response calculation. The resulting r_j is pooled across eligible counter-days and counter sections, and is applied as a common interval-specific response when reweighting station–season weather frequencies.

Using the same station–season group g and weather interval j as in the weather-frequency O/E analysis, the traffic-weighted weather frequency is

$$p_{jg}^* = \frac{p_{jg} r_j}{\sum_h p_{hg} r_h}, \quad (3.4)$$

where p_{jg}^* is the reweighted proportion for interval j in station–season group g , p_{jg} is its original weather proportion, and r_j is the allocated observed-to-expected traffic ratio. The index h runs over all weather intervals to normalise the weights.

Expected counts are then recalculated as $A_g p_{jg}^*$, where A_g is the group accident total. Accident bins use weather at the recorded accident time, whereas r_j is based on a 24-hour traffic total repeated across the day’s daytime weather composition. Daily counters do not observe hourly traffic. The 2019–2024 response is also extrapolated to the 2007–2025 primary population. The result is therefore an approximate traffic-reweighted O/E adjustment rather than a directly observed traffic-adjusted O/E.

3.3.4 Vehicle-kilometres travelled (VKT) analysis

The main traffic rate is restricted to 2019–2024, when daily counter data are available. Eligible accidents must be rural injury accidents occurring between 07:00 and 24:00, have a counter section for the registered road and year, be successfully located on that section, have usable accident-time wind and gust within 20 km and five minutes, and have a traffic record on the accident date. For this analysis, the 20 km weather distance is measured from the counter-section location rather than from the accident coordinate. The same five-minute time rule applies as in the weather-frequency O/E analysis and can select the adjacent observation across midnight. The denominator is broader than the accident set: it uses every eligible recorded counter-section day with a nonnegative traffic count and rural road length, including days without an accident. Estimated exposure, in vehicle-kilometres travelled (VKT), is

$$\hat{V}_{cdb} = C_{cd} L_{c,y(d)} p_{s(c),m(d),b}, \quad (3.5)$$

where $\hat{V}_{cdb}$ is exposure for counter section c , date d , and weather interval b ; C_{cd} is the observed 24-hour traffic count; $L_{c,y(d)}$ is rural section length in kilometres for calendar year $y(d)$; $m(d)$ is the calendar month; and $s(c)$ is the weather station assigned to section c . The proportion $p_{s(c),m(d),b}$ is the 07:00–24:00 weather frequency for that station, month and interval. The weather frequencies are pooled across 2007–2025 to provide stable station–month proportions, while both traffic and accident eligibility remain restricted to 2019–2024. The full daily count is allocated among weather intervals using the 07:00–24:00 station–month weather proportions. These proportions sum to one and therefore conserve total daily exposure. Accidents are classified separately by observed event-time weather within 20 km and five minutes.

Because accident eligibility and weather frequencies cover 07:00–24:00 but traffic counts cover 24 hours, absolute rates use allocated daily VKT rather than directly observed daytime VKT. Interpretation therefore emphasises contrasts between weather intervals rather than the absolute rate level.

The reported rate is

$$R_b = \frac{O_b}{\sum_c \sum_d \hat{V}_{cdb}} 10^6, \quad (3.6)$$

where R_b is the accident rate per million estimated vehicle-km in weather interval b , O_b is its event-time accident count, and $\hat{V}_{cdb}$ is allocated exposure summed over eligible counter sections c and dates d . The factor 10^6 expresses the rate per million estimated vehicle-km.

Observed daily traffic multiplied by rural road length determines each eligible counter-day's total estimated VKT. Monthly weather frequency is used only to allocate that VKT among weather intervals. The same allocation is applied separately to mean wind, gust and temperature using each variable's pooled station-month frequency; the proportions are normalised within each variable so that allocated exposure sums to the same total daily VKT. Because traffic is available only as daily totals, the main analysis uses the pooled station-month weather distribution rather than the weather realised on each individual day. This preserves observed day-to-day traffic variation but does not reconstruct hourly exposure. Actual event-time weather classifies the accident numerator.

For the injury-component figures, both rates share the exposure denominator:

$$R_b^{\text{minor}} = \frac{O_b^{\text{minor}}}{\hat{V}_b} 10^6, \quad R_b^{\text{serious/fatal}} = \frac{O_b^{\text{serious/fatal}}}{\hat{V}_b} 10^6, \quad (3.7)$$

where b indexes weather intervals, O_b^{minor} and $O_b^{\text{serious/fatal}}$ are the respective accident counts, and $\hat{V}_b$ is total monthly-allocated exposure in that interval. The corresponding R_b terms are rates per million estimated vehicle-km. Minor injury means code 3; serious or fatal injury means codes 1–2. Their counts are disjoint, so their rates sum to the all-injury rate. Blue forms the bottom of each stack and red the top. This is a descriptive decomposition of the accident numerator, not a severity model. For seasonal panels, counts and exposure are summed within each study season before division; the seasons partition the All year result. In the seasonal rate figures, mean-wind intervals at or above 15 m/s and gust intervals at or above 20 m/s are combined by summing counts and exposure before division. All year retains the original intervals.

4 Results

Results follow the three analysis stages defined in Chapter 3: weather-frequency O/E, the approximate traffic correction, and the VKT analysis. Their estimates are not directly comparable because the analyses use different samples and denominators.

4.1 Study sample and coverage

Wind and temperature have identical match coverage in the rural injury accident sample: 6,259 of 6,414 accidents (97.6%) for each measure. The annual match percentages remain high throughout the study (Figure 4.2), so weather linkage does not restrict the primary analysis to only a small subset of study years.

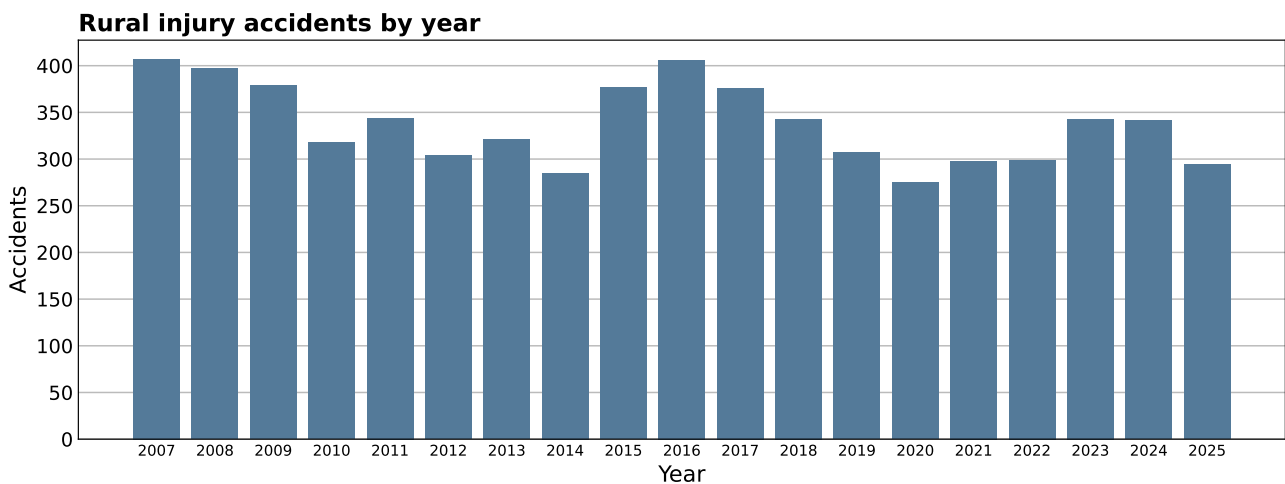


Figure 4.1: Annual rural injury accident counts, 2007–2025. Lower counts occur around 2010, 2014 and 2020. The 2010 and 2020 reductions coincide with the post-2008 economic downturn and COVID-19 period, respectively; counts alone cannot establish causation or separate travel volume from accident risk. The reason for the particularly low count in 2014 remains unclear.

The VKT analysis uses a smaller, counter-linked population. Table 4.1 follows selection to the final 694 accidents for the VKT analysis in Section 4.4. All 83 exclusions at the final step lack a counter-day record on the accident date; none is removed because its recorded daily count is zero.

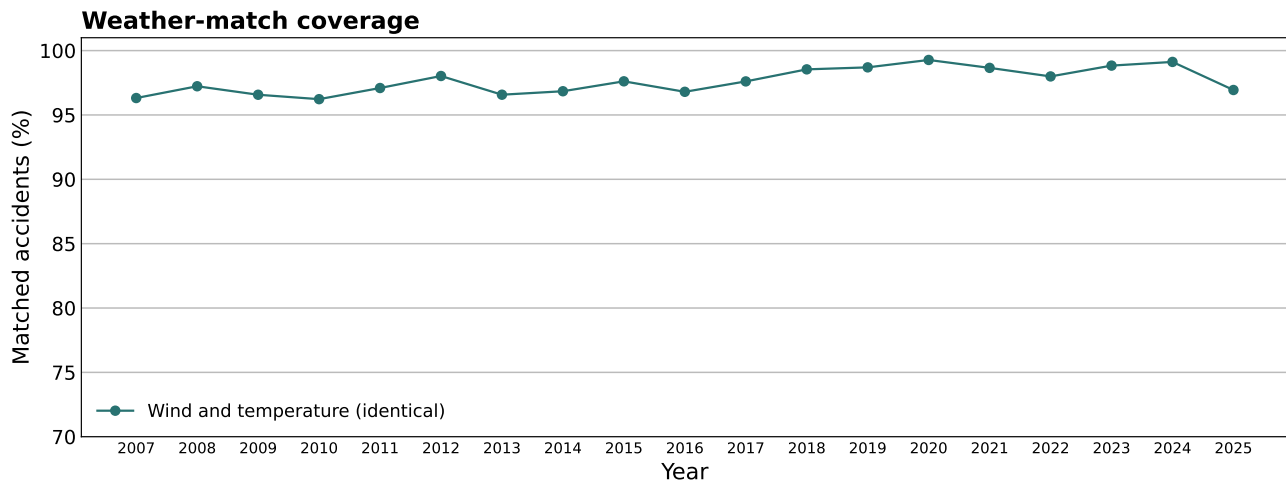


Figure 4.2: Annual weather-match coverage among rural injury accidents, 2007–2025. Matches require a qualifying observation within 20 km and five minutes; wind and temperature percentages coincide in each year.

Table 4.1: Selection for the VKT analysis. The first row includes urban and rural injury accidents; the rural restriction is applied in step (b). The final set contains minor, serious and fatal injury accidents. Removal percentages use the number entering each step.

Step	Removed (%)	Remaining
All injury accidents, 2007–2025	–	17,045
a) Restrict to 2019–2024	69.5%	5,196
b) Exclude urban accidents	64.1%	1,863
c) Require counter-section for road/year	53.4%	869
d) Require successful road-location assignment	2.4%	848
e) Exclude 00:00–07:00	7.9%	781
f) Require usable wind and gust (≤ 20 km, ± 5 min)	0.5%	777
g) Require daily traffic count	10.7%	694

Notes: The 21 road-location failures in step (d) comprise five records without usable road geometry, 15 located more than 100 m from their registered road, and one projected outside the counter sections. For step (f), the 20 km weather distance is measured from the counter-section location rather than the accident coordinate; valid mean wind and gust are required within five minutes of the recorded accident time.

4.2 Weather-frequency O/E analysis

Figures 4.3–4.5 show the primary weather-frequency comparisons for the full year and four study seasons. Blue and red represent the disjoint minor injury and serious or fatal injury groups, and $O/E = 1$ marks equality between observed and expected counts. These are comparisons with local weather frequency, not rates per vehicle-kilometre; sparse seasonal upper bins are interpreted descriptively. Throughout the O/E and VKT figure families, panel-specific y-axis limits aid readability; comparisons across panels should use axis values rather than relative bar height.

4.2.1 Mean wind speed and accident frequency

Figure 4.3 shows the main result: O/E is close to one at low and moderate mean wind speeds and rises in the upper intervals. At 20 m/s or above, the all-injury O/E is 2.15, and O/E also increases in the upper wind intervals in both injury groups. This means that strong wind is over-represented at accident times relative to how often it occurs locally; it does not yet account for traffic exposure.

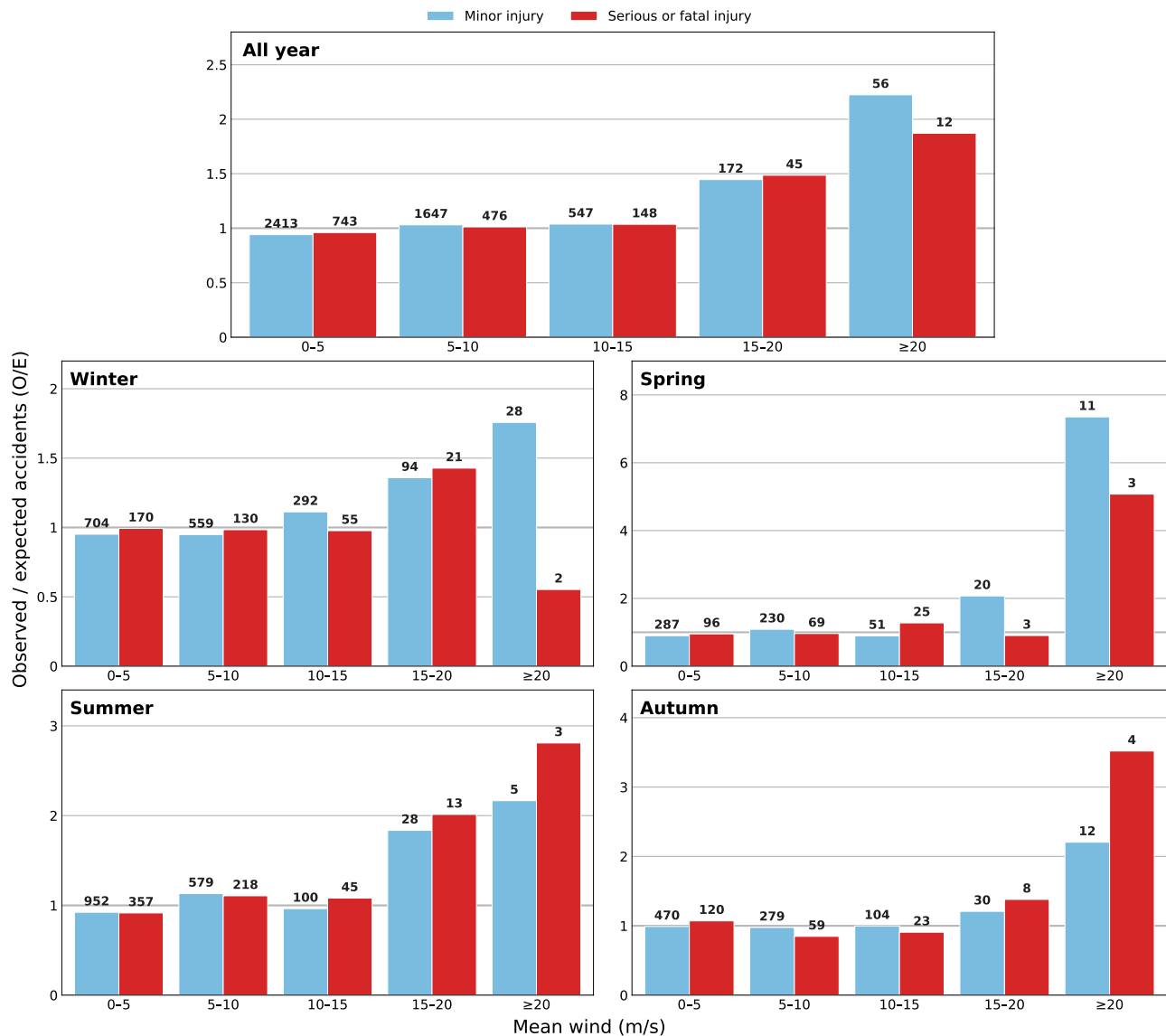


Figure 4.3: All year and seasonal mean-wind O/E analysis, 2007–2025. Blue and red show minor and serious or fatal injury; expected counts use local station–season weather frequency separately for each group. Labels give observed accident counts and the grey line marks O/E = 1.

The seasonal panels are less stable than the pooled result and do not support ranking seasons or injury groups. The 5 m/s intervals are descriptive bins, not physical wind thresholds.

4.2.2 Wind gust and accident frequency

Wind gust shows a similar upper-tail pattern in Figure 4.4, with all-injury O/E 3.34 at gusts of at least 30 m/s. Because gust and mean wind come from the same observation, the gust result supports the same high-wind pattern but is not independent evidence.

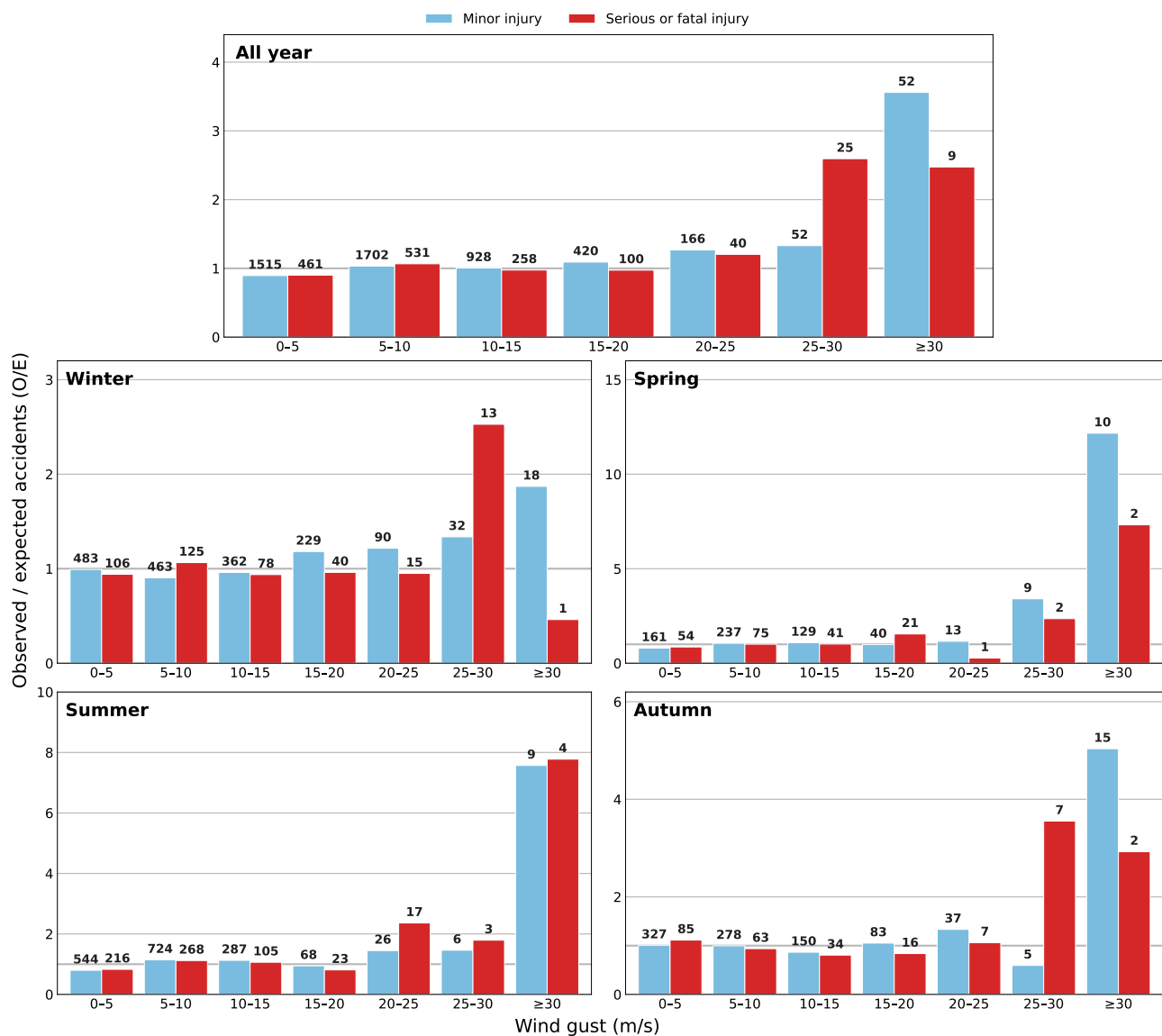


Figure 4.4: All year and seasonal wind-gust O/E analysis. Gust is from the same 10-minute observation as mean wind, not a daily maximum. Blue and red show minor and serious or fatal injury; labels give observed counts and the grey line marks $O/E = 1$.

4.2.3 Temperature and accident frequency

Temperature shows a non-monotonic pattern in Figure 4.5: O/E is lower at moderate temperatures and higher in some colder and warmer intervals. The all-injury O/E is 1.22 immediately below freezing, falls to 0.71 at 3–6°C, and rises to 1.60 at temperatures of at least 12°C. The coldest group, below –6°C, has O/E 0.84; thus the pattern is not a uniform cold-weather excess. Seasonal warm-end components are smaller and less stable than the pooled comparison.

Adding year to the station–season standardisation changes the selected O/E estimates only slightly. The station–season standardisation is therefore retained as the primary weather-frequency O/E comparison.

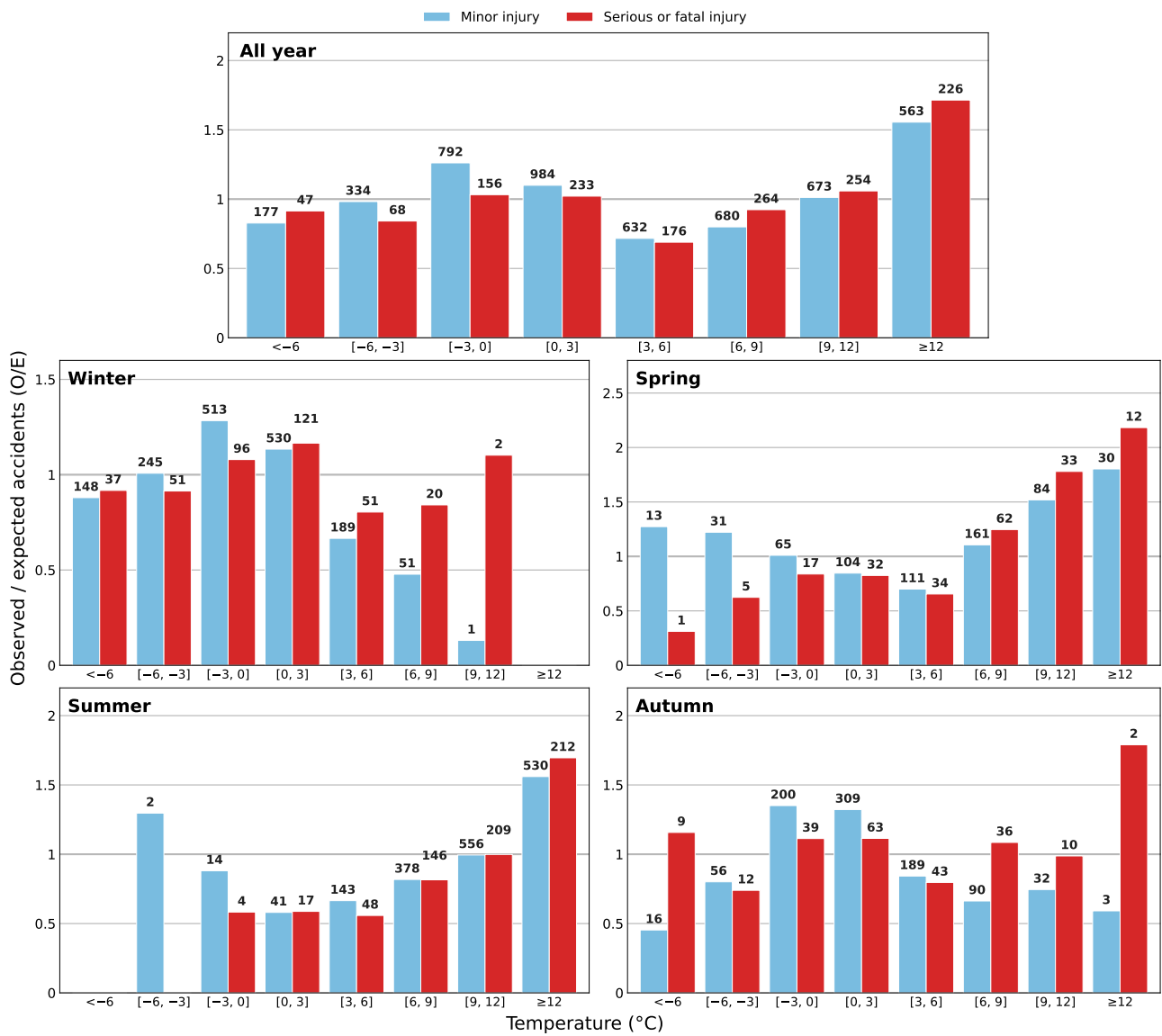


Figure 4.5: All year and seasonal temperature O/E analysis, standardised within matched temperature station and study season. Blue bars show minor injury and red bars show serious or fatal injury; labels above bars are observed accident counts. The grey reference line is $O/E = 1$.

4.2.4 Joint wind and temperature pattern

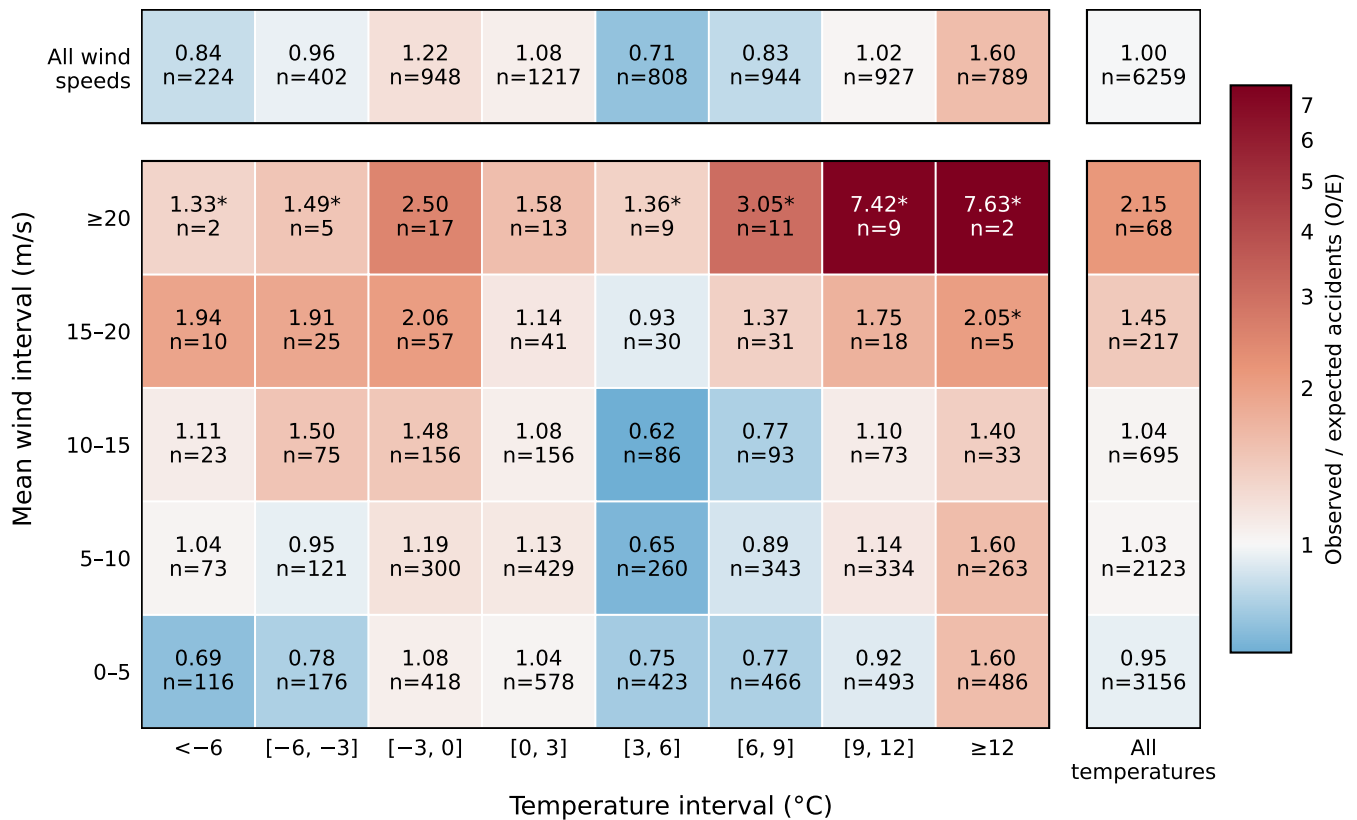


Figure 4.6: O/E analysis of accident frequency according to both temperature and wind speed. Note that, independent of wind speed, the rate is significantly increased in warm weather. Also temperatures around zero are associated with increased accident rate at all wind speeds. Interestingly, calm and very cold weather is among the safest conditions. Also calm weather is a little bit riskier than a moderate breeze. Perhaps good weather makes drivers speed too much?

Figure 4.6 uses the same five wind and eight temperature intervals as the marginal figures. The all-temperature O/E rises from 0.95 at 0–5 m/s to 1.45 at 15–20 m/s and 2.15 at 20 m/s or above. Within the 15–20 m/s row, adequately supported cells range from 0.93 at 3–6°C to 2.06 immediately below freezing. The all-wind temperature margin is 1.22 immediately below freezing and 1.60 at 12°C or above, but 0.71 at 3–6°C.

Six of the eight cells at wind speeds of at least 20 m/s have limited support. The largest O/E (7.63) occurs at 20 m/s or above and at least 12°C, but represents only 2 observed accidents versus 0.26 expected. That extreme cell should not be read as evidence of a combined wind–temperature effect.

4.3 Approximate traffic-corrected O/E analysis

4.3.1 Traffic behaviour and the baseline limitation

Figure 4.7 shows the traffic response underlying the correction. Values below 100% indicate less traffic than the counter's calendar expectation, not fewer accidents. The upper wind and gust intervals show reductions. The temperature response is much smaller across the displayed intervals. Daily traffic is 22.3% below its calendar expectation at mean wind of at least 20 m/s and 21.5% below expectation at gusts of at least 30 m/s. These reductions describe the denominator response used by the correction, not an accident-rate change.

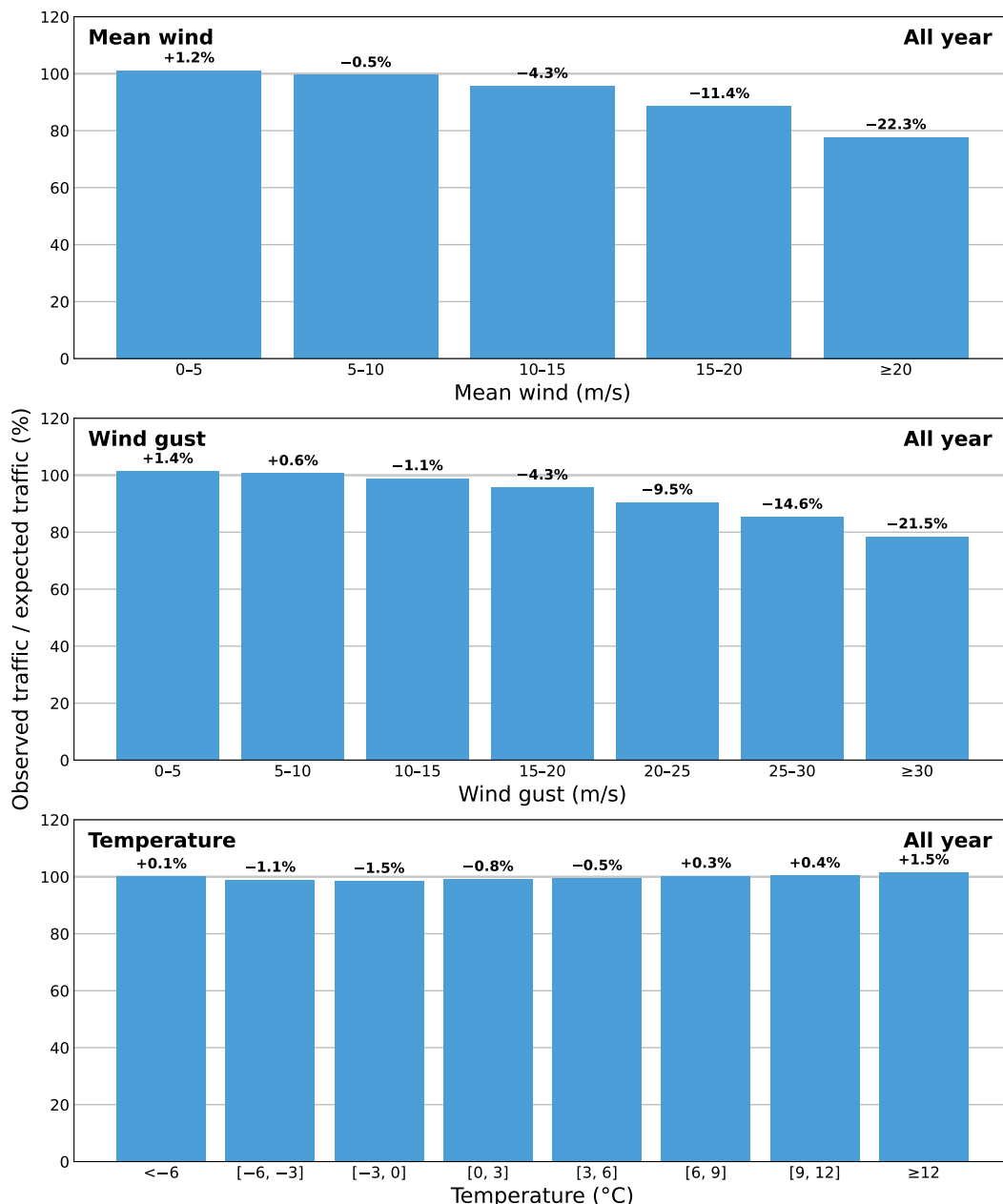


Figure 4.7: Observed daily traffic relative to expected traffic at counter sections, 2019–2024, pooled across seasons. The expectation uses the same section, year, month and weekday among positive-count days. Counts are allocated by observed 07:00–24:00 weather-bin minutes; bars show the ratio of pooled allocated observed to expected traffic. The line marks 100%; labels show percentage changes, not accident counts.

The response is then used as an approximate correction, assuming that the selected counters are informative about the wider rural network.

4.3.2 Full-period correction and injury-group patterns

The approximate correction reweights the primary station–season denominator using the 2019–2024 counter response; observed accident counts remain unchanged.

Figures 4.8 and 4.9 show the effect directly: upper-bin O/E rises from 2.15 to 2.71 for mean wind and from 3.34 to 4.20 for gust. Accounting for lower traffic therefore strengthens the high-wind contrast. In other words, the observed reduction in traffic during strong wind does not explain the primary O/E excess; within this approximate correction, it makes the contrast larger. The full-period correction is reported only for the All-year comparison.

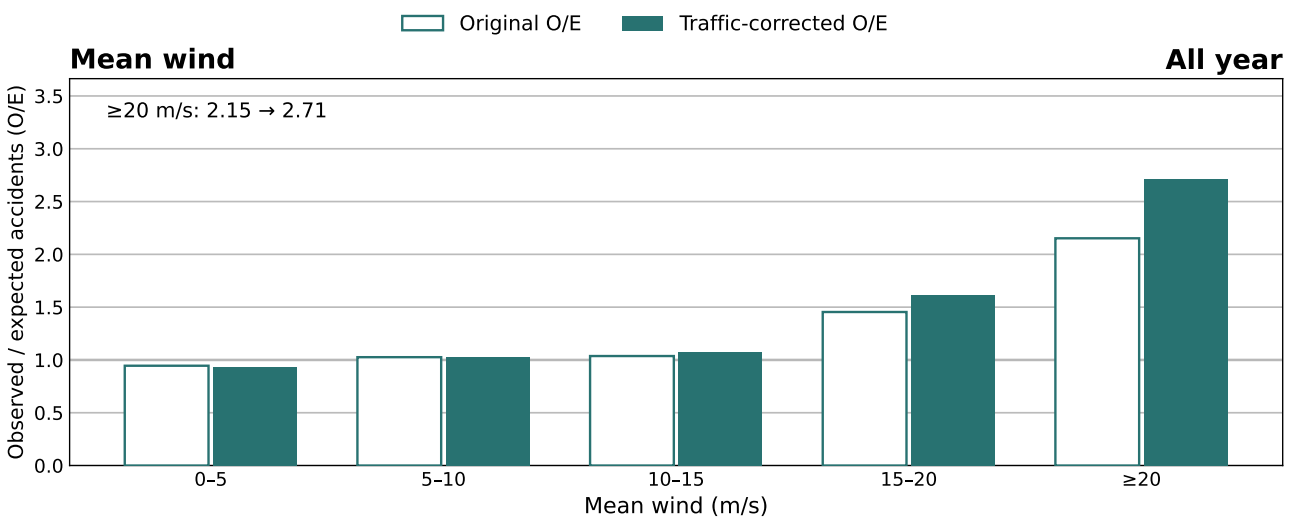


Figure 4.8: Original and approximately traffic-corrected mean-wind O/E analysis, 2007–2025, All year. Outlined bars use weather frequency and teal bars use the 2019–2024 counter response; the line marks $O/E = 1$.

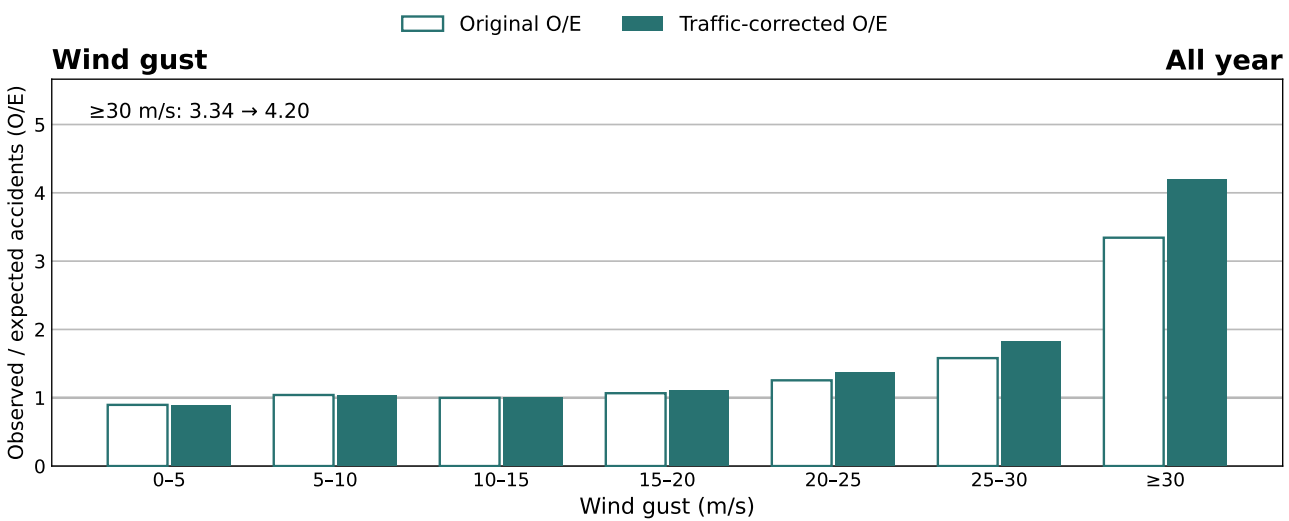


Figure 4.9: Original and approximately traffic-corrected wind-gust O/E analysis, 2007–2025, All year. Outlined and teal bars use the original and reweighted expected counts, respectively; the line marks $O/E = 1$.

Figure 4.10 compares corrected mean-wind, gust and temperature O/E separately for minor and serious or fatal injury. Both injury groups show an upper-wind increase; temperature varies non-linearly across the displayed intervals. The highest wind and gust categories have limited counts, particularly for serious or fatal injury. These injury-group differences are descriptive and are not a test of severity.

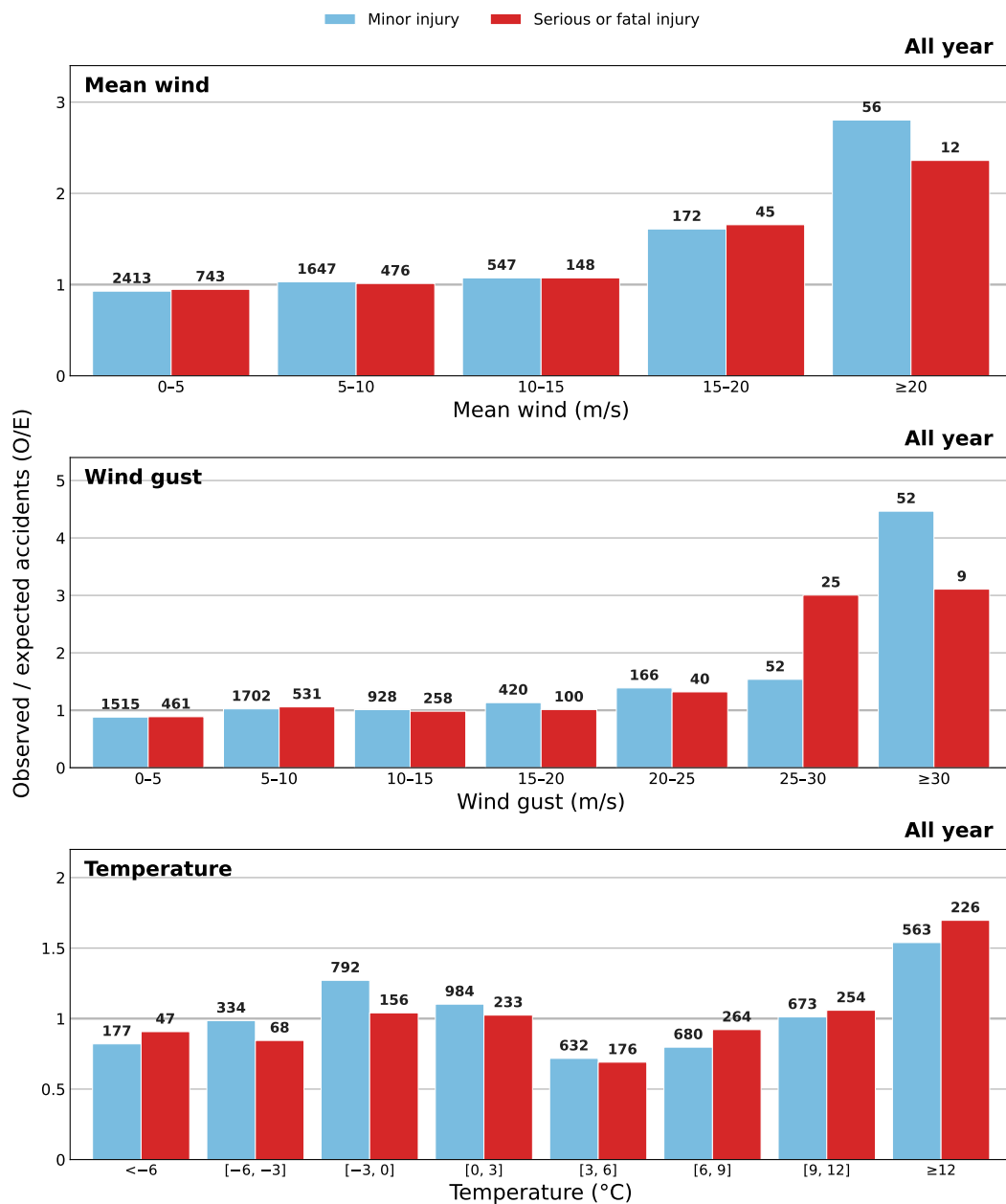


Figure 4.10: Approximately traffic-corrected mean-wind, gust and temperature O/E analysis by injury group, 2007–2025, All year. Expected counts are reweighted using the 2019–2024 counter response; labels give observed counts and the line marks $O/E = 1$.

4.3.3 Temperature after traffic correction

The full-period temperature correction leaves the non-linear pattern largely intact (Figure 4.11). All-injury O/E is 1.23 at $-3-0^{\circ}\text{C}$, 0.71 at $3-6^{\circ}\text{C}$ and 1.58 at or above 12°C , compared with 1.22, 0.71 and 1.60 before correction. The coldest interval, below -6°C , remains below one (0.84). Thus the data support an excess just below freezing, not a universal excess throughout the cold range.

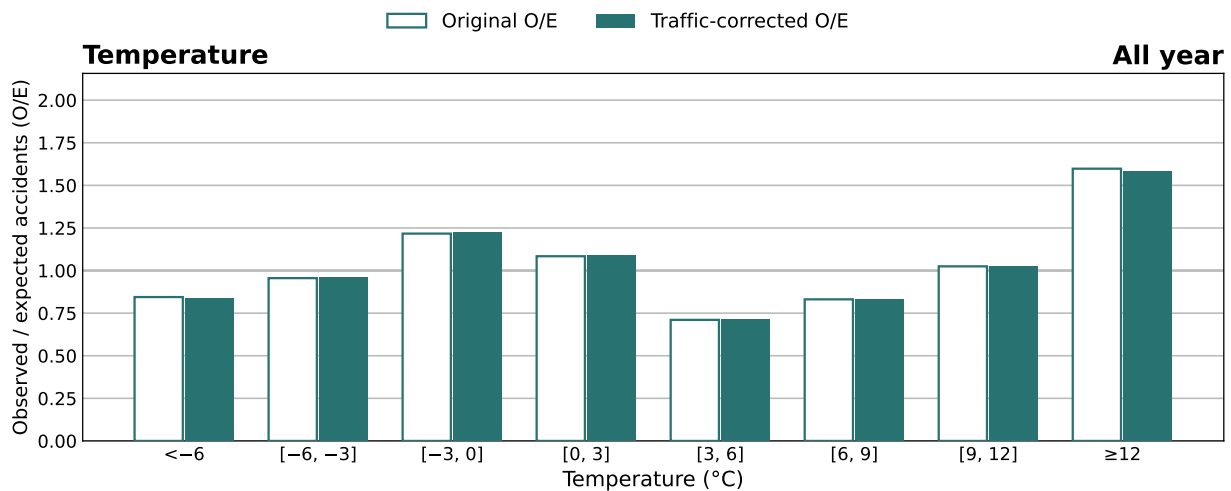


Figure 4.11: All-injury temperature O/E before and after approximate traffic correction, 2007–2025. The response measured in 2019–2024 reweights each station–season weather distribution.

4.4 Vehicle-kilometres travelled (VKT) analysis

The third stage estimates accident rates in the smaller counter-linked sample. For each eligible counter-day, observed daily traffic and rural road length determine total estimated VKT, which is then allocated among weather intervals using pooled monthly weather frequencies at the assigned station. Table 4.2 gives the resulting wind and gust counts, exposure and rates. Temperature uses the same pooled station–month allocation and the existing temperature intervals.

Rates rise markedly in the upper wind intervals. At mean wind of at least 20 m/s, the rate is 5.83 times the 0–5 m/s rate; for gusts of at least 30 m/s, the contrast is 7.78. Unlike the O/E analyses, these are rates per estimated travel exposure. The persistence of the upper-wind increase therefore shows that the wind association remains when an estimated VKT denominator is introduced. Only 11 accidents contribute to each upper interval, so the magnitude of these contrasts is uncertain, and the selective counter coverage limits their generalisability.

4.4.1 All year rates

Figure 4.12 shows the pooled wind and gust rates. The light-blue and red components share the same denominator, so their stack height is the all-injury rate; labels give contributing accident counts.

Table 4.2: VKT wind and gust rates, 2019–2024. Exposure is the full observed daily traffic count times rural section length, allocated by the assigned station’s pooled 2007–2025 calendar-month weather frequency for 07:00–24:00.

Measure	Interval (m/s)	Observed	Million estimated vehicle-km	Accidents per million estimated vehicle-km
Mean wind	0–5	344	3011.5	0.114
Mean wind	5–10	210	1963.5	0.107
Mean wind	10–15	96	539.9	0.178
Mean wind	15–20	33	98.9	0.334
Mean wind	≥20	11	16.5	0.667
Wind gust	0–5	205	1840.9	0.111
Wind gust	5–10	238	2133.5	0.112
Wind gust	10–15	129	1085.1	0.119
Wind gust	15–20	70	400.3	0.175
Wind gust	20–25	29	123.3	0.235
Wind gust	25–30	12	34.5	0.348
Wind gust	≥30	11	12.7	0.866

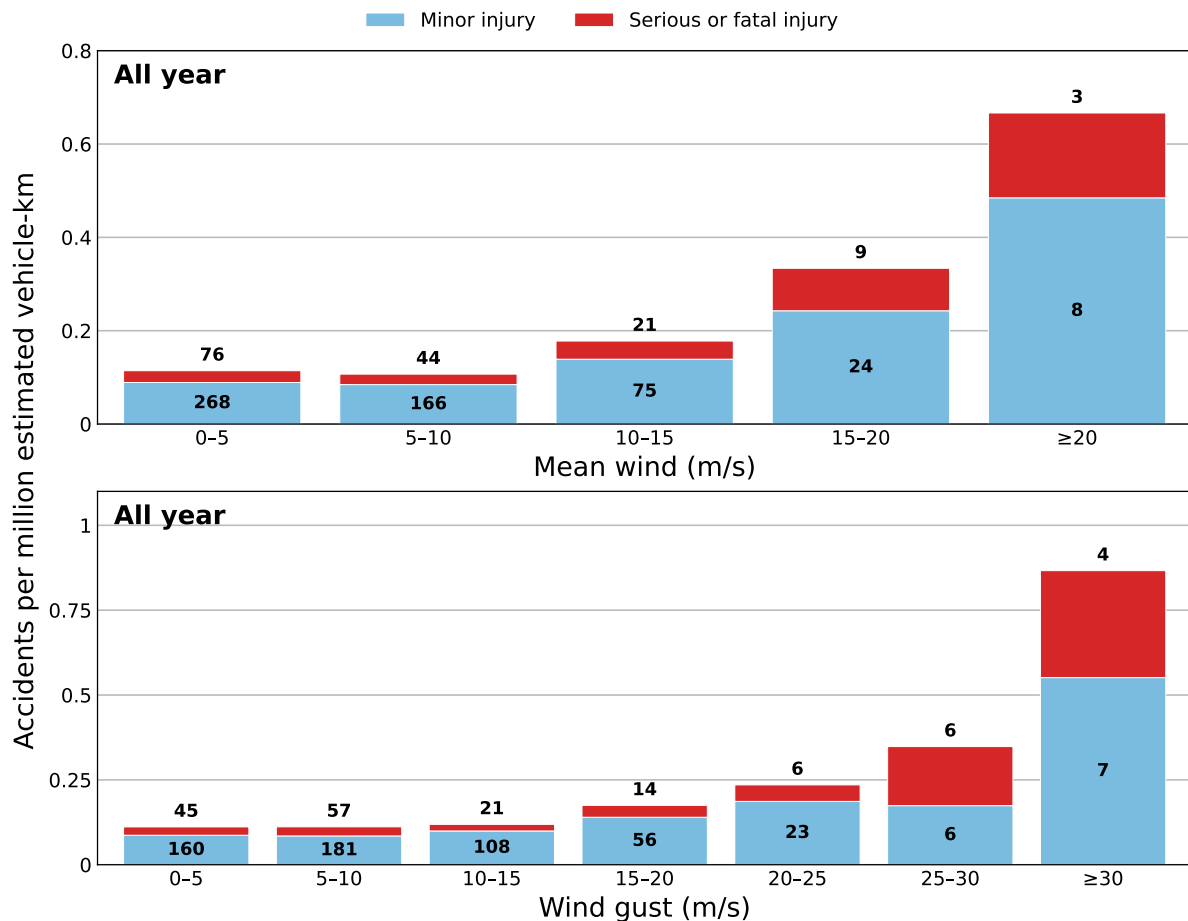


Figure 4.12: Rural injury accident rates per million estimated vehicle-km, 2019–2024, using the VKT denominator defined in Section 3.3.4. Event-time wind and gust classify accidents. Light blue and red show minor and serious or fatal injury; positive labels give accident counts.

Temperature. The VKT extension uses the same 694 event-time observations and 533,649 counter-days. Allocated exposure sums to 5.630 billion vehicle-kilometres for each variable. All-injury rates decrease from 0.282 below -6°C to 0.067 at $9\text{--}12^{\circ}\text{C}$, then rise to 0.099 at or above 12°C . The warm-end rate is 1.48 times the $9\text{--}12^{\circ}\text{C}$ rate, but remains well below the cold rates. This supports a modest warm-end upturn, not the same magnitude or reference comparison as the temperature O/E result. The eight temperature intervals contain 40, 61, 120, 134, 84, 95, 79 and 81 accidents, respectively; seasonal components are smaller.

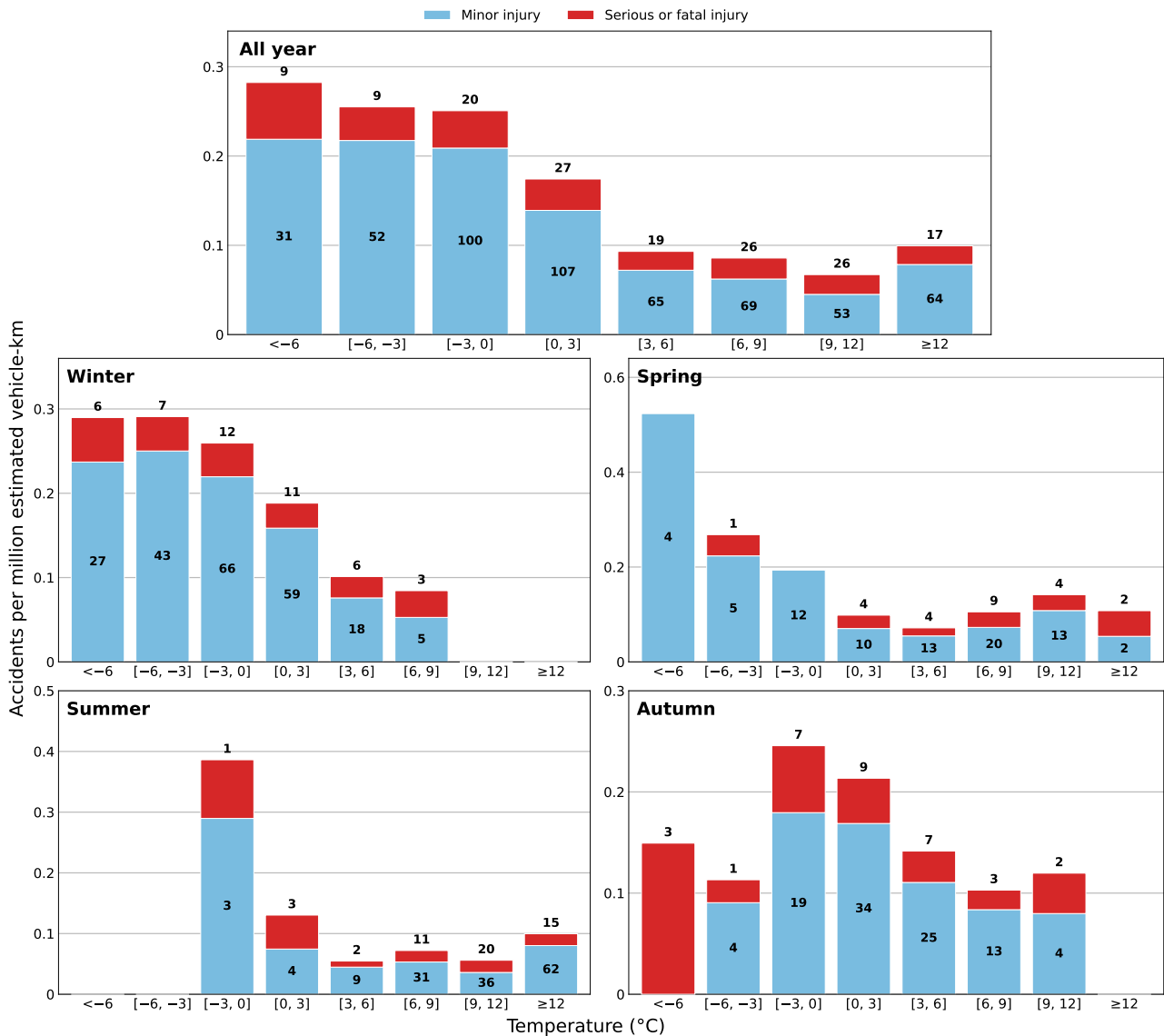


Figure 4.13: Rural injury accidents per million estimated vehicle-kilometres by temperature, 2019–2024. Exposure uses pooled station–month temperature frequencies; event-time temperature classifies accidents. Labels give minor and serious/fatal counts, and the stack gives their combined rate.

4.4.2 Mean wind by season

Figure 4.14 shows mean-wind rates by season, with the seasonal upper tail combined at 15 m/s. Rates rise in the upper interval in several seasonal panels. Upper-to-lowest-interval rate ratios are 1.93, 7.19, 4.76 and 1.25 in winter, spring, summer and autumn, respectively. The upper groups contain 22, 10, 7 and 5 accidents. These descriptive ratios summarise the seasonal panels; limited counts and different

seasonal exposure do not support a stable ranking of seasons.

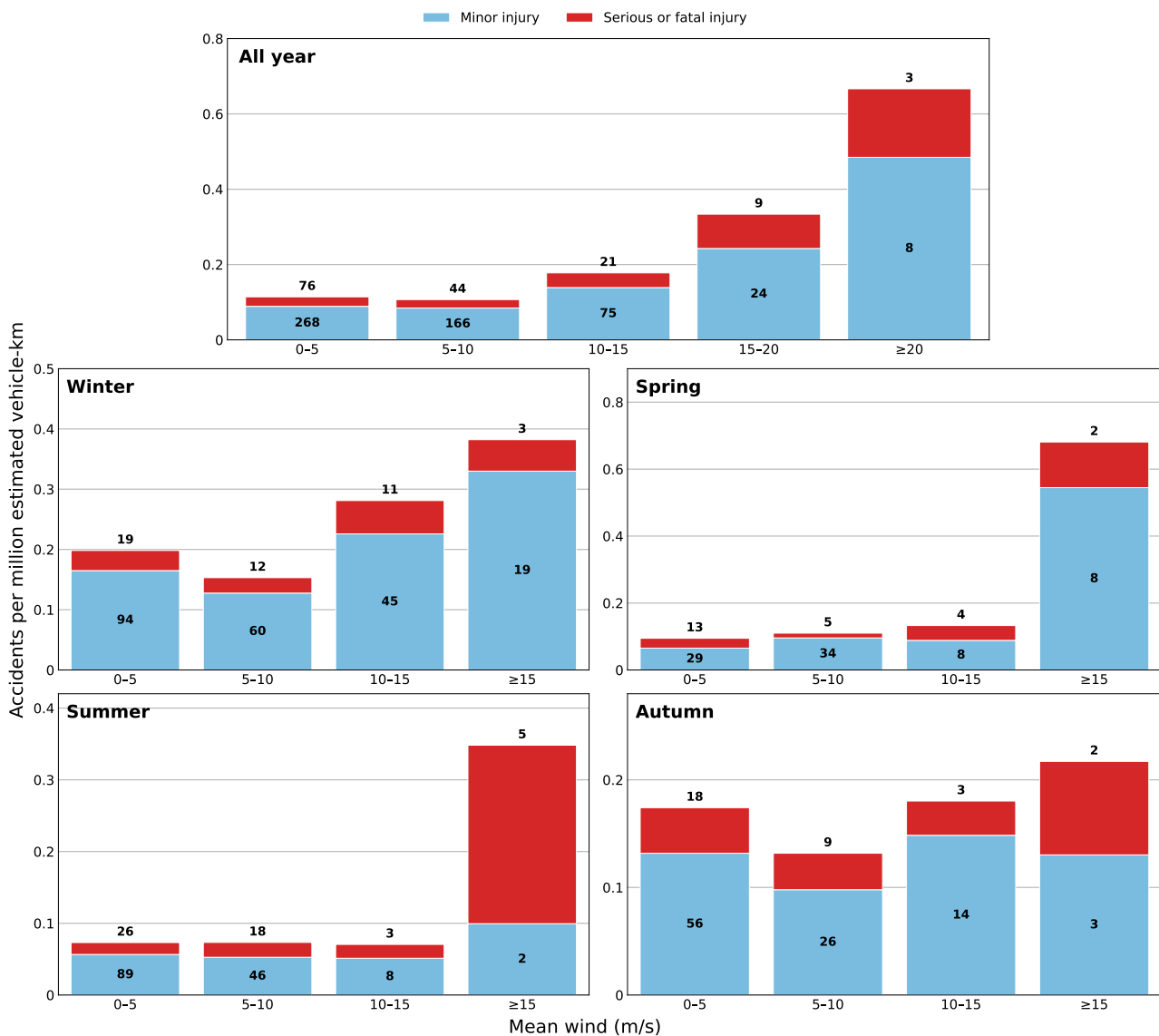


Figure 4.14: Rural injury accident rates by mean wind, 2019–2024, using the same VKT denominator as Figure 4.12. All year retains original intervals; seasonal tails combine intervals from 15 m/s. Light blue and red show minor and serious or fatal injury; positive labels give counts.

4.4.3 Wind gust by season

Figure 4.15 shows the corresponding gust rates, with the seasonal upper tail combined at 20 m/s. Each season shows higher rates in its upper interval than in its lowest interval. Upper-to-lowest-interval rate ratios are 1.74, 3.59, 4.26 and 1.42 in winter, spring, summer and autumn, respectively. The upper groups contain 26, 8, 10 and 8 accidents. These descriptive ratios summarise the seasonal panels; limited counts and different seasonal exposure do not support a stable ranking of seasons.

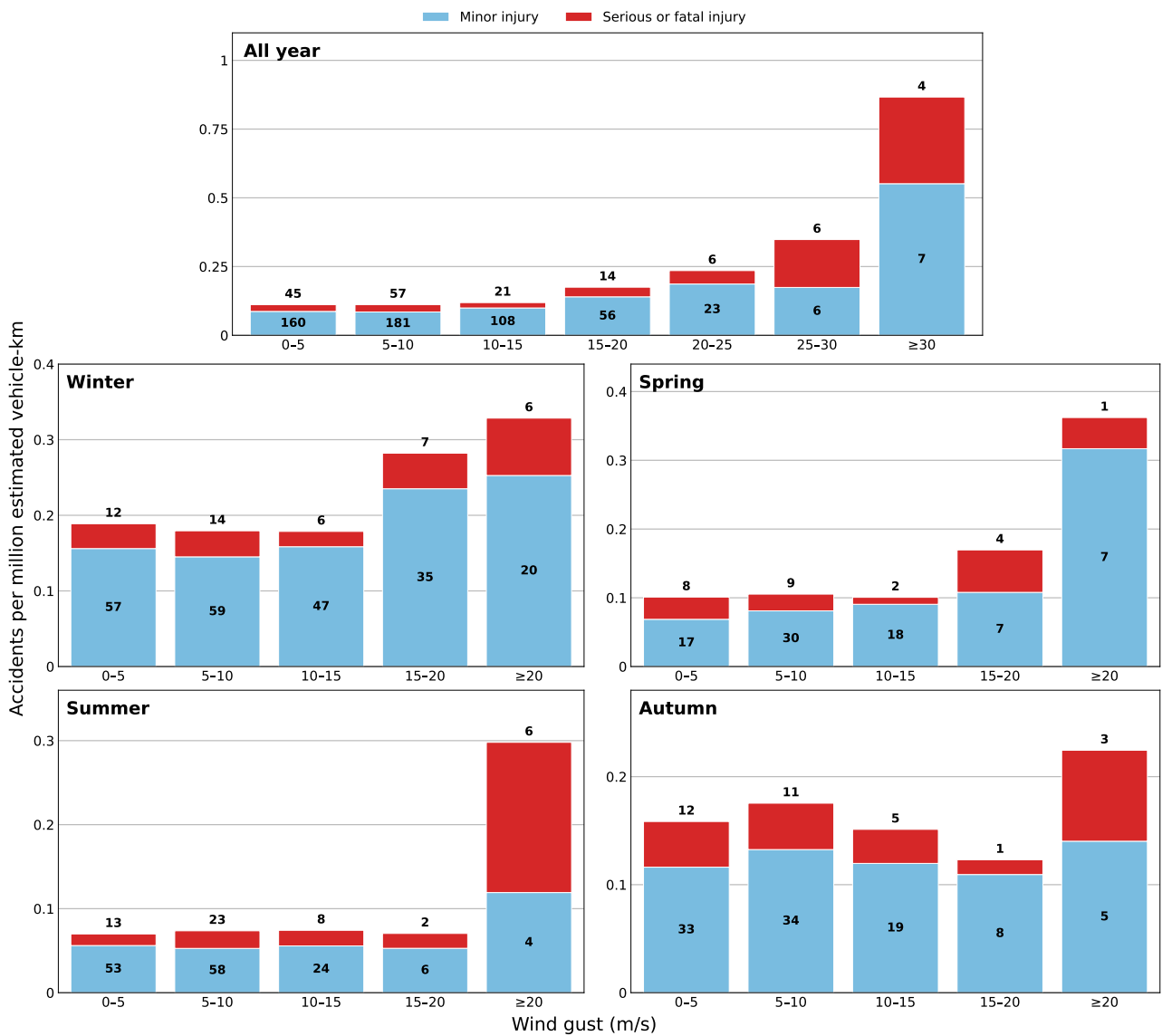


Figure 4.15: Rural injury accident rates by gust, 2019–2024, using the same VKT denominator as Figure 4.12. All year retains original intervals; seasonal tails combine intervals from 20 m/s. Light blue and red show minor and serious or fatal injury; positive labels give counts.

4.4.4 Comparison with traffic-corrected O/E

Table 4.3 summarises the primary mean-wind and gust results after rescaling each method to its own low-wind reference. The displayed ratios share an upper-to-reference interpretation, but the underlying samples and denominators still differ.

Temperature further illustrates why the methods should not be equated: the O/E analyses show a pronounced warm-end increase, whereas the VKT warm-end upturn is smaller and the highest VKT rates occur in cold conditions.

The estimates differ because they answer different questions. Weather-frequency O/E compares accident occurrence with local weather time. Its approximate correction reweights that time using the relative traffic response at selected counters, extrapolating 2019–2024 behaviour to the broader 2007–2025 accident population. VKT instead divides accidents by estimated travel in a smaller counter-linked 2019–2024 sample. The contributing roads, periods and exposure weights differ; identical ratio notation does not make the estimands identical.

Table 4.3: Mean-wind and gust results. Ratios compare the upper interval with 0–5 m/s. Both O/E methods use 2007–2025 weather-matched rural injury accidents; the correction uses 2019–2024 counter data. VKT uses 2019–2024 accidents on counter-covered sections.

Parameter	Method	Comparison (m/s)	Ratio
Mean wind	Weather-frequency O/E analysis	≥ 20 vs 0–5	2.28
Mean wind	Approximate traffic-corrected O/E analysis	≥ 20 vs 0–5	2.91
Mean wind	VKT analysis	≥ 20 vs 0–5	5.83
Wind gust	Weather-frequency O/E analysis	≥ 30 vs 0–5	3.73
Wind gust	Approximate traffic-corrected O/E analysis	≥ 30 vs 0–5	4.75
Wind gust	VKT analysis	≥ 30 vs 0–5	7.78

The pooled VKT comparison also depends on where and when travel occurs. Windy and calm vehicle-kilometres need not come from the same mix of counter sections, years and seasons. If groups contributing more windy travel have higher accident rates overall, pooling can enlarge the wind ratio even without an equally large increase within each group. Counter placement on hazardous roads alone would not do this: those roads must contribute a different share of windy than of calm travel.

An exploratory diagnostic uses the same 694 VKT accidents but calculates expected counts within each counter-section-year-season group from its distribution of estimated VKT. The resulting upper-to-0–5 m/s ratios are 3.43 for mean wind and 5.16 for gust, compared with pooled VKT rate ratios of 5.83 and 7.78. Differences in exposure composition account arithmetically for factors of about 1.70 and 1.51 in the pooled ratios. This does not establish that particular roads are intrinsically dangerous, nor does it remove the wind association within groups.

Other differences remain. Restricting corrected O/E to 2019–2024 gives ratios of 2.84 for mean wind and 5.95 for gust; the change of years therefore explains little of the mean-wind gap but some of the gust gap. The VKT upper bins each contain only 11 accidents, making their ratios sensitive to sampling variation. Moreover, VKT allocates daily traffic across wind intervals using pooled station-month weather frequencies, not observed travel during the actual windy hours. A comparison using one common accident sample, with uncertainty estimates and more direct traffic exposure, is needed to determine the contribution of each mechanism.

4.4.5 Descriptive full-period road-rate comparison

This descriptive road-level calculation is separate from the main counter-linked VKT analysis above. The full-period ranking aggregates injury accidents and annual estimated rural VKT by registered road number. Annual exposure is annual average daily traffic times mapped rural length times calendar days, the year-round counterpart of the winter and summer daily-traffic construction. Only accidents linked to road-section years with positive estimated exposure enter the numerator. Of the 6,414 rural injury accidents, 5,498 link to such section-years and 916 are excluded from the ranking numerator. The full unfiltered CSV retains every road with exposure, including zero-accident roads; it does not assume that a road is intrinsically dangerous.

The road ordering could be biased if the 916 accidents without a link to road-section years with positive estimated exposure are concentrated on particular roads. Whole-road aggregation does not identify hazardous locations within roads. This comparison is not weather-specific and does not attribute differences between roads to wind.

The unfiltered top ten contain eight roads with only one accident. Publishing those as a stable top ten

would be misleading. For display, Table 4.4 restricts the ranking to roads with at least 20 linked injury accidents. Under a simple Poisson count approximation this limits count-only relative standard error to about 22%; it does not address exposure error and is not a significance threshold. The unfiltered ranking remains the audit source. Sparse counts, changing identifiers and incomplete historical geometry still limit interpretation.

Table 4.4: Roads with the highest estimated injury-accident rates, 2007–2025. Only roads with at least 20 linked injury accidents are shown; the rates are approximate.

Road	Road name	Injury accidents	Million estimated vehicle-km	Accidents per million estimated vehicle-km
59	Laxárdalsvegur	20	35	0.58
93	Seyðisfjarðarvegur	46	91	0.51
429	Sandgerðisvegur	37	96	0.38
42	Krýsuvíkurvegur	38	101	0.37
39	Þrengslavegur	51	155	0.33
92	Norðfjarðarvegur	69	211	0.33
61	Djúpvegur	149	464	0.32
76	Siglufjarðarvegur	80	253	0.32
829	Eyjafjarðarbraut eystri	22	71	0.31
56	Vatnaleið	21	70	0.30

5 Discussion

5.1 Interpretation

Across the analyses, the strongest contrasts occur in the upper wind intervals. The primary weather-frequency result shows that strong wind is over-represented at rural injury accident times. The approximate traffic correction shows that lower traffic during strong wind does not explain that over-representation, and the VKT analysis retains the upper-wind increase when estimated travel exposure is introduced. The analyses therefore agree on the direction of the wind association, but they do not estimate the same quantity: their denominators differ and the VKT analysis uses a smaller counter-linked sample.

The correction remains approximate: the 2019–2024 counter response is extrapolated to the 2007–2025 population, and hourly traffic is unavailable.

The VKT analysis also shows higher rates in the upper wind intervals, but its rate ratios are larger than the O/E contrasts. Those magnitudes are uncertain: only 11 accidents fall in each highest wind or gust interval, counter coverage is selective, and pooled rates depend partly on which road sections contribute exposure. The smaller within-counter-section–year–season contrasts suggest that road and exposure composition explain part of the difference between pooled VKT rate ratios and locally standardised O/E, without implying that particular roads are intrinsically more dangerous.

Pooled station–month allocation conserves daily VKT but approximates the weather distribution of travel. Gust shows a similar upper-tail pattern, although mean wind and gust come from the same observations. The temperature association is non-linear: the O/E analyses show an excess just below freezing and at the warm end, whereas VKT rates are highest in cold intervals and rise again above the 9–12°C minimum. The coldest O/E interval is below one, and the warm-end VKT rate remains well below the cold rates. The three analyses therefore do not support identical temperature contrasts.

The finer joint grid retains the upper-wind and non-linear temperature patterns visible in the marginal figures. Crossing the two variables leaves most cells at 20 m/s or above sparsely supported, however; differences between individual temperature-specific wind ratios remain descriptive and do not establish interaction.

Several alternative mechanisms could contribute to the joint pattern. Drivers may reduce speed, avoid travel or change routes when hazards are apparent, while warnings and closures may change who remains on the road. These are hypotheses, not evidence that drivers compensate when wind and near-freezing temperatures coincide. Warm-weather excess could reflect tourism, motorcycles, recreational travel, vehicle mix, road choice or speed rather than a direct temperature effect. Correlated precipitation, ice, road-surface state and geography may also contribute. Behavioural adaptation, travel avoidance, closures, changing vehicle composition and direct physical effects of weather cannot be distinguished with the available data.

The full-period seasonal exposure summary similarly shows why raw counts can mislead: winter contained fewer injury accidents than summer, but the estimated injury-accident rate per rural VKT was about 1.92 times the summer rate because estimated winter travel was much lower.

The wind findings are consistent with earlier accident-frequency comparisons (Edwards, 1994) and with engineering evidence that crosswind can impair vehicle stability (Baker, 1986, 1994). The study links a long Icelandic accident series to local weather frequency and examines the wind association using several traffic-exposure constructions. The non-monotonic temperature results are also compatible with previous evidence of range- and season-dependent temperature associations (Lee et al., 2014; Zhan et al., 2020), while the observed traffic reduction in strong wind agrees with recorder-based evidence that adverse weather changes travel demand (Sathiaraj et al., 2018).

5.2 Strengths

The primary analysis combines a long accident series with 10-minute weather observations screened using explicit validity rules and standardises expected counts within station and study season rather than against one national weather distribution. High match coverage and explicit reporting of weather exclusions make the O/E calculation transparent.

The two traffic analyses address the main limitation of the weather-frequency denominator in different ways. The approximate correction reweights expected counts using the observed counter response, whereas the VKT analysis uses observed daily traffic, rural road length, and accident-free counter-days to estimate exposure. The scripted workflow generates thesis tables and validates sample counts, matching rules and exposure conservation.

5.3 Limitations

The accident register does not capture every traffic injury. Because it is based primarily on police-reported accidents, minor injuries are particularly susceptible to underreporting; health institutions and insurers record additional cases that do not enter the official register (ITA, 2026b).

Traffic exposure is the main limitation. The primary O/E uses weather time rather than vehicle-kilometres, and the approximate correction extrapolates a 2019–2024 response from selected counters to the broader 2007–2025 population. Because traffic is observed only as daily totals, the traffic analyses cannot reconstruct the hourly exposure corresponding to accident-time weather.

The VKT analysis is also selective: it is limited to rural counter sections with usable traffic, road geometry, and weather in 2019–2024. Its monthly weather frequencies are pooled over 2007–2025 for stability, while traffic and accidents remain restricted to the counter era. Only a few accidents occur in the highest wind and gust intervals, so the large rate ratios should not be treated as precise estimates of magnitude. A counter represents exposure on a road section and does not show that an accident-involved vehicle passed that counter.

Weather stations are proxies for road conditions. A station may be up to 20 km from an accident or counter location, and Icelandic terrain can produce large variation over shorter distances. The fixed 2020–2024 urban-boundary dataset is applied to the full accident period, so historical settlement-boundary changes are not represented. Unusual traffic, road-surface conditions, visibility and closures are not controlled. More generally, unmeasured road characteristics, vehicle type, speed, driver behaviour, and weather-related road management may contribute to the observed patterns. The study therefore supports associations and exposure contrasts, not causal effects of wind.

5.4 Future work

Future research should first improve the measurement of travel exposure. Hourly traffic counts would align traffic with accident-time weather and distinguish daytime travel from the 24-hour totals currently allocated to accidents eligible during 07:00–24:00. Broader counter coverage, verified coordinates and stable road identifiers would improve linkage and reduce dependence on a selective sample. Counter-placement audits should compare monitored and unmonitored roads, while independent estimates from traffic surveys or other travel records could validate the VKT denominator. These steps would help establish whether weather contrasts persist under more directly observed exposure.

Weather measurements should also better represent conditions along the road. Roadside sensors or evaluated higher-resolution meteorological models could reduce dependence on a station some distance from an accident. Wind direction relative to road orientation would distinguish crosswind, headwind and tailwind, while gust duration and short-term variability could reveal information lost in a single observation. Precipitation, visibility, road-surface temperature, snow and ice should be linked to maintenance records, warnings and closures. Validation against roadside observations would be needed before treating modelled weather as an improvement over the current measurements.

Behaviour and exposure response require separate observation. Speed measurements and repeated traffic records could test whether drivers slow down, avoid travel, change routes or respond to warnings as conditions deteriorate. Combining these records with closure times would help distinguish voluntary travel avoidance from enforced restrictions. With sufficient observations, later work could examine whether responses differ when several hazards occur together. The present analysis cannot distinguish behavioural adaptation from reduced exposure, and a lower accident count under combined hazards would not by itself demonstrate safer driving. Comparisons should therefore measure both travel volume and behaviour.

The highest-priority extension would test whether the upper-wind association is concentrated in single-vehicle crashes, road departures, overturning or loss-of-control events, after auditing accident-type coding. This would connect the population-level association more directly to the crosswind mechanisms discussed in the Background. Separate person- and vehicle-level extracts supplied for 2007–2024 may support subgroup analyses, subject to completeness and coding audits. Those extracts alone exclude 2025; including that year would require harmonising the separate 2025 deliveries already available. Vehicle-class codes may support comparisons among passenger cars, vans, heavy goods vehicles, buses and motorcycles. Without traffic exposure by vehicle class, these would describe accident composition under different weather conditions, not vehicle-class-specific accident risk. Road curvature, gradient, elevation and orientation could distinguish exposed coastal roads, mountain passes and sheltered routes. Multilevel or spatial models could separate within-road weather variation from between-road differences, with reliable identifiers and exposure and allowance for dependence within roads and weather episodes.

Statistical development should quantify uncertainty in O/E and VKT contrasts and assess sensitivity to weather bins, matching distance, temporal tolerance, season definitions and exposure-allocation assumptions. Resampling or other uncertainty methods should preserve relevant station, road and temporal dependence. Flexible non-linear models could examine temperature without relying solely on grouped intervals. Formal wind–temperature interaction should be considered only with adequate joint sample size and support across the exposure range. Validation in later years would test whether estimated patterns generalise beyond the study period, while comparison with simpler models would assess whether added complexity improves interpretation or predictive performance.

Around freezing, measured temperature should be compared with police-recorded road-surface or road-condition fields, if obtained and sufficiently complete, to assess whether the elevation immediately below freezing coincides with recorded ice or snow. At the warm end, the available person and vehicle data may help examine motorcycle involvement, injured road-user age or road-user type. They contain no confirmed tourism indicator. Such comparisons would assess composition rather than assume that ambient temperature changes driver care.

The accompanying *Töflulýsingar í slysaskrá* describes the source database schema, not the completeness of delivered fields. Fields such as recorded cause, reported weather and actual speed according to the party may be subjective or incomplete and are not included in the delivered extracts. Future use should begin with missingness, coding-consistency and temporal-coverage audits. Following independent validation, findings might inform targeted warnings, monitoring, counter placement or road management. The present analysis is not ready for operational prediction, and more covariates or more complex models would not by themselves establish causation.

6 Conclusion

Strong wind was the clearest and most consistent weather association in the study. Rural injury accidents were substantially over-represented during strong wind relative to how often those conditions occurred locally: at mean wind speeds of at least 20 m/s, O/E was 2.15. In the second step, observed lower daily traffic in strong wind did not explain away the pattern; the approximate correction strengthened the upper-wind contrast. The third step, the smaller counter-linked VKT analysis, also showed higher rates in the strongest wind conditions, although their magnitude remains uncertain because counter coverage is selective and only 11 accidents occur in each highest wind or gust category.

Temperature showed a non-linear pattern. Both O/E analyses found elevated occurrence just below freezing and at temperatures of at least 12°C, with lower relative occurrence at moderate temperatures; the coldest interval was not elevated above local weather frequency. VKT rates were highest in cold conditions and rose again at the warm end relative to the 9–12°C minimum, but this warm-end increase was smaller than the O/E contrast. The retained analyses support non-linearity, not one common temperature effect. The warm-temperature increase should be treated as an association requiring further study rather than evidence of a direct causal effect.

These findings describe associations, not causation or a direct national risk estimate. Exposure data remain the main limitation. Verified hourly traffic, broader counter coverage, stable locations and identifiers, and better measurement of road-level exposure composition would improve the comparison. Future work should incorporate vehicle class, wind direction relative to the road, road geometry, road-surface and closure information, and investigate the unexpected warm-temperature pattern.

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